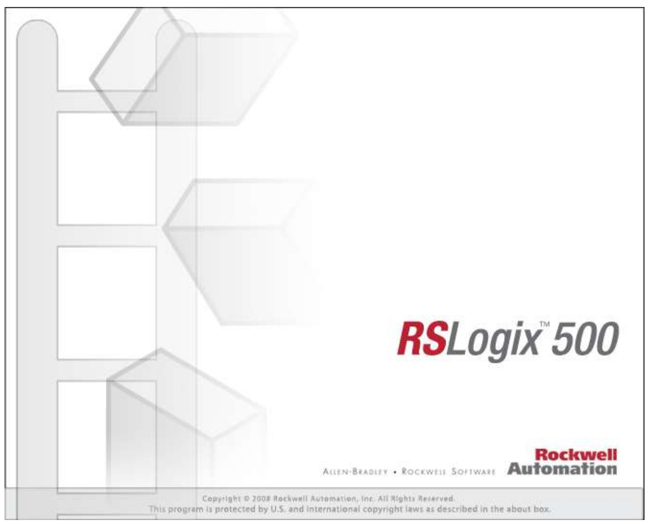
RSLogix Micro Project Report



Processor Information

Processor Type: Bul.1763   MicroLogix 1100 Series B

Processor Name: UNTITLED

Total Memory Used: 816 Instruction Words Used - 148 Data Table Words Used

Total Memory Left: 5840 Instruction Words Left

Program Files: 11

Data Files: 9

Program ID: d5f4

I/O Configuration

0	Bul.1763	MicroLogix 1100 Series B
1	1762-IF4	Analog 4 Chan. Input
2		
3		
4		

Channel Configuration

CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex

CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Edit Resource/Owner Timeout: 60
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Passthru Link ID: 1
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Write Protected: No
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Comms Servicing Selection: Yes
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Message Servicing Selection: Yes
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex 1st AWA Append Character: \d
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex 2nd AWA Append Character: \a

Source ID: 1 (decimal)
Baud: 19200
Parity: NONE
Control Line : No Handshaking
Error Detection: CRC
Embedded Responses: Auto Detect
Duplicate Packet Detect: Yes
ACK Timeout(x20 ms): 50
NAK Retries: 3
ENQ Retries: 3

CHANNEL 1 (SYSTEM) - Driver: Ethernet

CHANNEL 1 (SYSTEM) - Driver: Ethernet Edit Resource/Owner Timeout: 60
CHANNEL 1 (SYSTEM) - Driver: Ethernet Passthru Link ID: 1
CHANNEL 1 (SYSTEM) - Driver: Ethernet Write Protected: No
CHANNEL 1 (SYSTEM) - Driver: Ethernet Comms Servicing Selection: Yes
CHANNEL 1 (SYSTEM) - Driver: Ethernet Message Servicing Selection: Yes

Hardware Address: 00:00:00:00:00:00
IP Address: 0.0.0.0
Subnet Mask: 0.0.0.0
Gateway Address: 0.0.0.0
Msg Connection Timeout (x 1mS): 15000
Msg Reply Timeout (x mS): 3000
Inactivity Timeout (x Min): 30
Bootp Enable: No
Dhcp Enable No
SNMP Enable: No
HTTP Enable: Yes
Auto Negotiate Enable: Yes
Port Speed Enable: 10/100 Mbps Full Duplex/Half Duplex
Contact:
Location:

Program File List

Name	Number	Type	Rungs	Debug	Bytes
[SYSTEM]	0	SYS	0	No	0
	1	SYS	0	No	0
MAIN	2	LADDER	9	No	75
IO	3	LADDER	13	No	444
HOA	4	LADDER	36	No	1228
CONTROLS	5	LADDER	16	No	665
RUNTIME	6	LADDER	6	No	258
BACKWASH	7	LADDER	16	No	437
ALARM	8	LADDER	17	No	547
SIMULATION	9	LADDER	9	No	304
SETTOAUTO	10	LADDER	3	No	213

Data File List

Name	Number	Type	Scope	Debug	Words	Elements	Last
OUTPUT	0	O	Global	No	12	4	O:3
INPUT	1	I	Global	No	39	13	I:12
STATUS	2	S	Global	No	0	66	S:65
BINARY	3	B	Global	No	9	9	B3:8
TIMER	4	T	Global	No	33	11	T4:10
COUNTER	5	C	Global	No	3	1	C5:0
CONTROL	6	R	Global	No	3	1	R6:0
INTEGER	7	N	Global	No	19	19	N7:18
FLOAT	8	F	Global	No	30	15	F8:14

Rung	Comment	Instruction	Description	SBR File Number
0000	I/O	JSR	Jump To Subroutine	U:3
0001	HOA	JSR	Jump To Subroutine	U:4
0002	CONTROLS	JSR	Jump To Subroutine	U:5
0003	RUNTIME	JSR	Jump To Subroutine	U:6
0004	BACKWASH	JSR	Jump To Subroutine	U:7
0005	ALARM	JSR	Jump To Subroutine	U:8
0006	SIMULATION	JSR	Jump To Subroutine	U:9
0007	SETTOAUTO	JSR	Jump To Subroutine	U:10
0008		END		

CREATING INPUT BITS FOR THE OUTPUTS. THIS WILL MAKE THE PROCESS OF MODIFYING THE OUTPUT ADDRESSES QUICKER IN CASE THE OUTPUTS ARE CHANGED ON THE PLC. INSTEAD OF HAVING TO CHANGE AN OUTPUT ADDRESS IN MULTIPLE PLACES IN THE PROGRAM, IT WILL ONLY BE CHANGED HERE ONCE.
THE INPUT BITS WILL BE USED TO ACTIVATE THE OUTPUTS IN OTHER FILES IN THE PROGRAM, AND THOSE WILL REMAIN UNCHANGED SINCE THEY WILL NOT CHANGE WITH SWITCHING OUTPUTS ON THE PLC.

Rung	Input description	Input symbol	Input address	Input bit	Output description	Output address	Output bit	Output module
0000	Pump	P1	B3:0	7	Pump OP	O:0	0	Bul.1763
0001	Flow switch FS-1	FS1	B3:0	0	FS-1	O:0	1	Bul.1763
0002	SV1	SV1	B3:0	1	SV1	O:0	2	Bul.1763
0003	SV2	SV2	B3:0	2	SV2	O:0	3	Bul.1763
0004	SV3	SV3	B3:0	3	SV3	O:0	4	Bul.1763
0005	SV4 ip	SV4	B3:0	4	SV5	O:0	5	Bul.1763
0006	SV5	SV5	B3:0	5	SV5	O:0	6	Bul.1763
0007	SV6	SV6	B3:0	6	SV6	O:0	7	Bul.1763
0008	DISCHARGE VALVE	DISCHARGE_VALVE	B3:7	12	DISCHARGE VALVE	O:0	8	Bul.1763

SCALING ALL ANALOG INPUTS.

0009

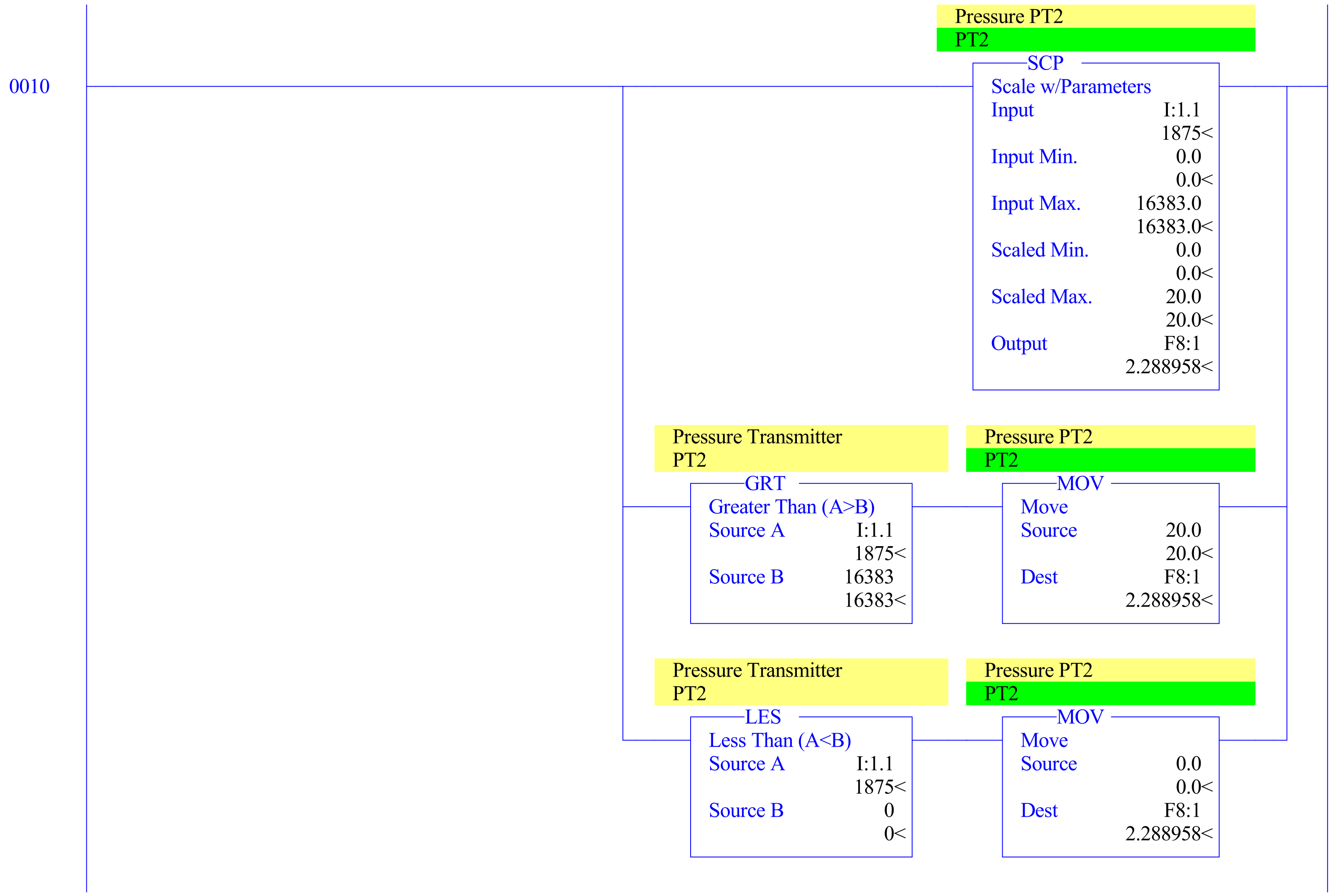
Pressure PT1
PT1
SCP
Scale w/Parameters
Input I:1.0
50<
Input Min. 0.0
0.0<
Input Max. 16383.0
16383.0<
Scaled Min. 0.0
0.0<
Scaled Max. 20.0
20.0<
Output F8:0
0.06103888<

Pressure transmitter
PT1
GRT
Greater Than (A>B)
Source A I:1.0
50<
Source B 16383
16383<

Pressure PT1
PT1
MOV
Move
Source 20.0
20.0<
Dest F8:0
0.06103888<

Pressure transmitter
PT1
LES
Less Than (A<B)
Source A I:1.0
50<
Source B 0
0<

Pressure PT1
PT1
MOV
Move
Source 0.0
0.0<
Dest F8:0
0.06103888<


0010
Pressure PT2
PT2
SCP
Scale w/Parameters
Input I:1.1
1875<
Input Min. 0.0
0.0<
Input Max. 16383.0
16383.0<
Scaled Min. 0.0
0.0<
Scaled Max. 20.0
20.0<
Output F8:1
2.288958<
Pressure Transmitter
PT2
GRT
Greater Than (A>B)
Source A I:1.1
1875<
Source B 16383
16383<
Pressure PT2
PT2
MOV
Move
Source 20.0
20.0<
Dest F8:1
2.288958<
Pressure Transmitter
PT2
LES
Less Than (A<B)
Source A I:1.1
1875<
Source B 0
0<
Pressure PT2
PT2
MOV
Move
Source 0.0
0.0<
Dest F8:1
2.288958<

0011

Tank Level Scaled
LT1
SCP
Scale w/Parameters
Input I:1.2
3000<
Input Min. 0.0
0.0<
Input Max. 16383.0
16383.0<
Scaled Min. 0.0
0.0<
Scaled Max. 100.0
100.0<
Output F8:3
18.31166<

Level Transmitter
GRT
Greater Than (A>B)
Source A I:1.2
3000<
Source B 16383
16383<

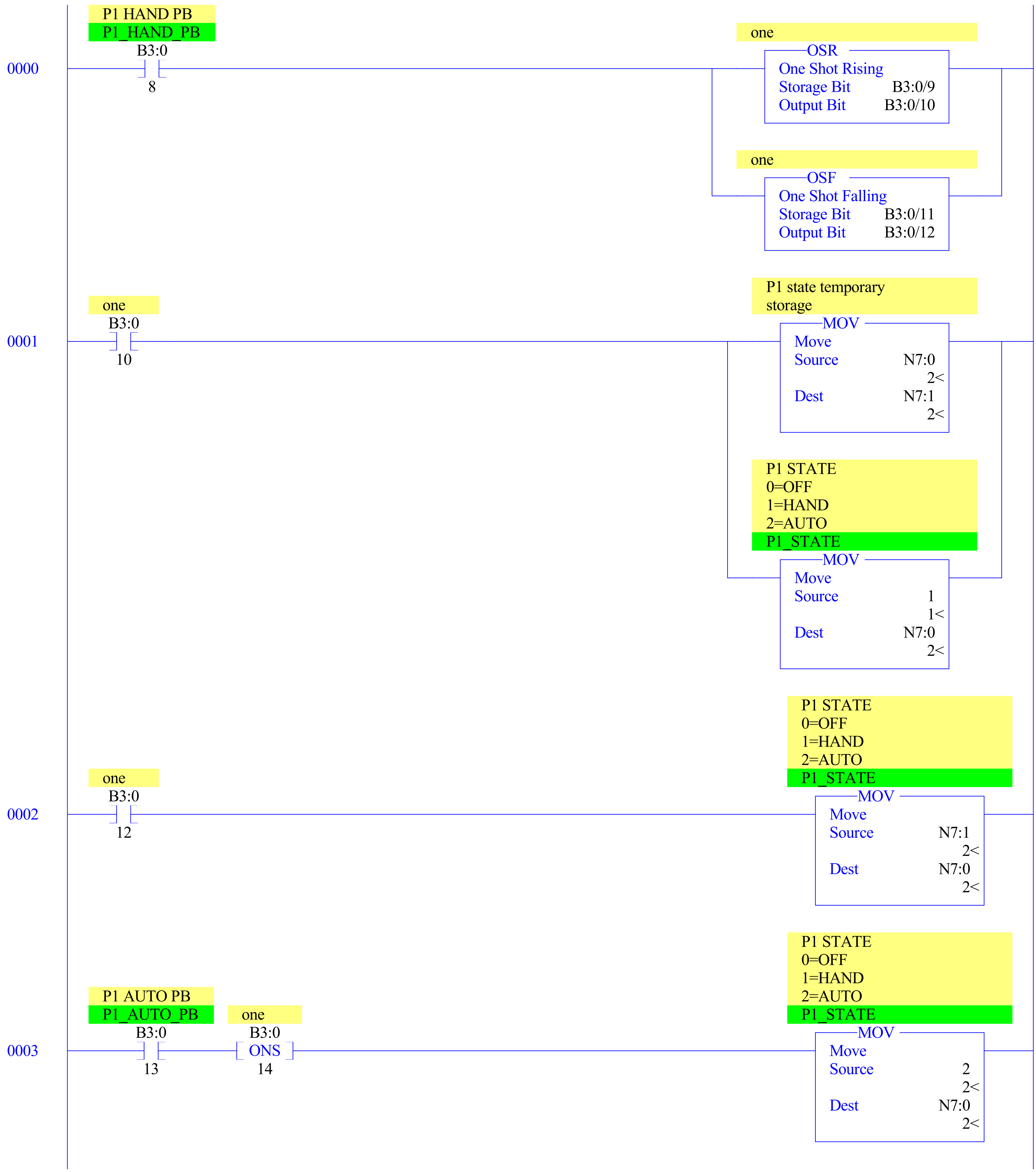
Tank Level Scaled
LT1
MOV
Move
Source 100.0
100.0<
Dest F8:3
18.31166<

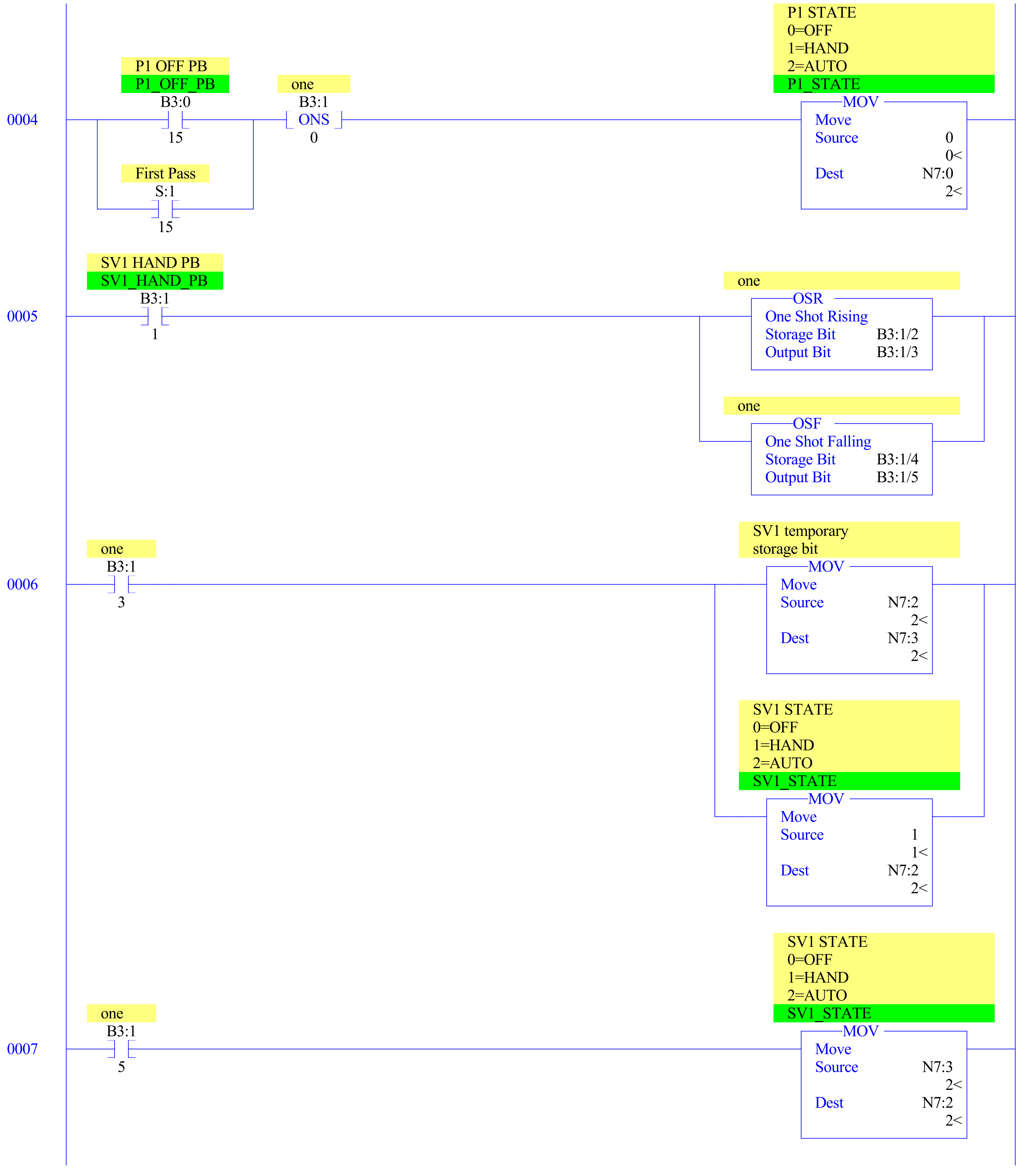
Level Transmitter
LES
Less Than (A<B)
Source A I:1.2
3000<
Source B 0
0<

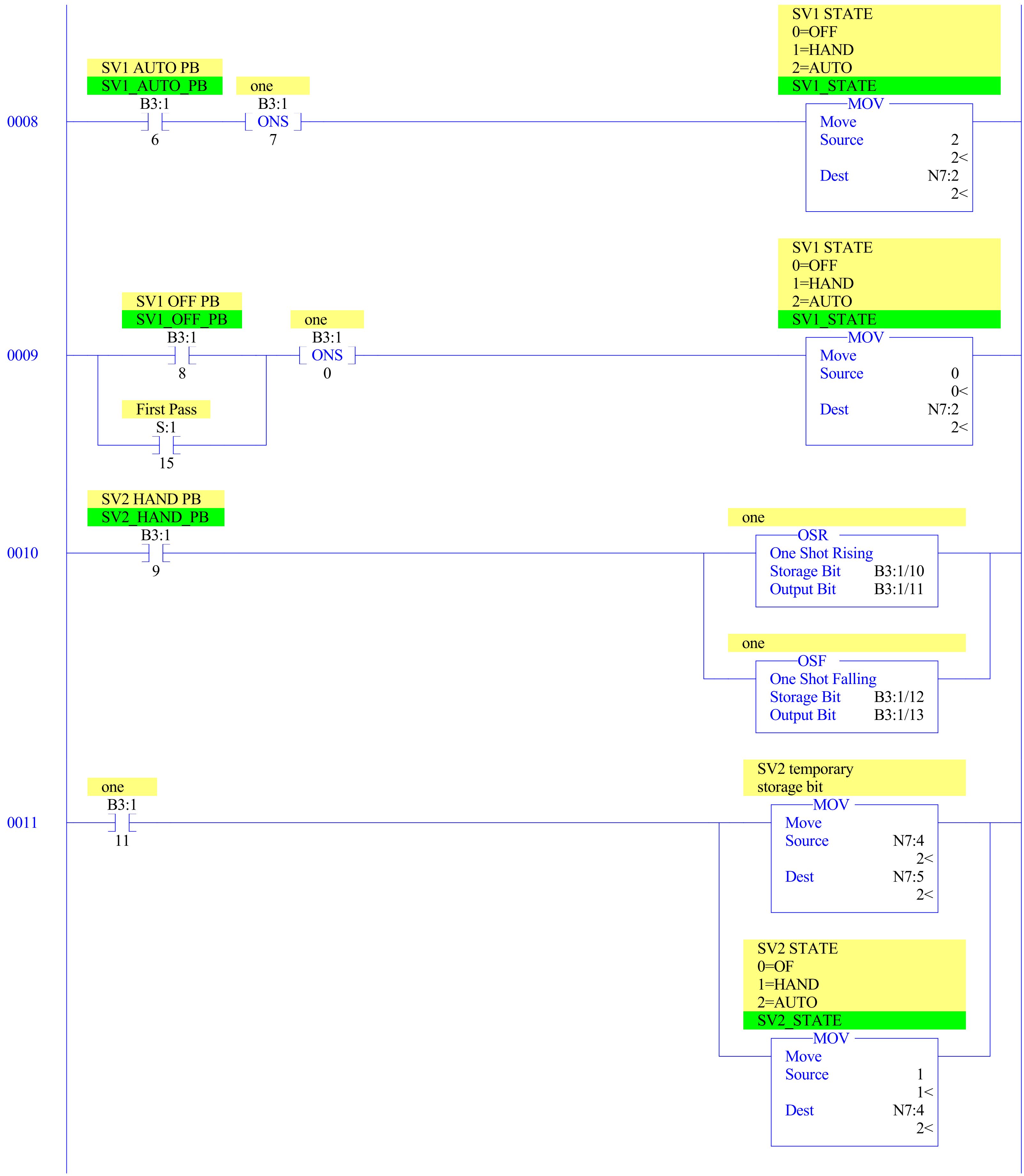
Tank Level Scaled
LT1
MOV
Move
Source 0.0
0.0<
Dest F8:3
18.31166<

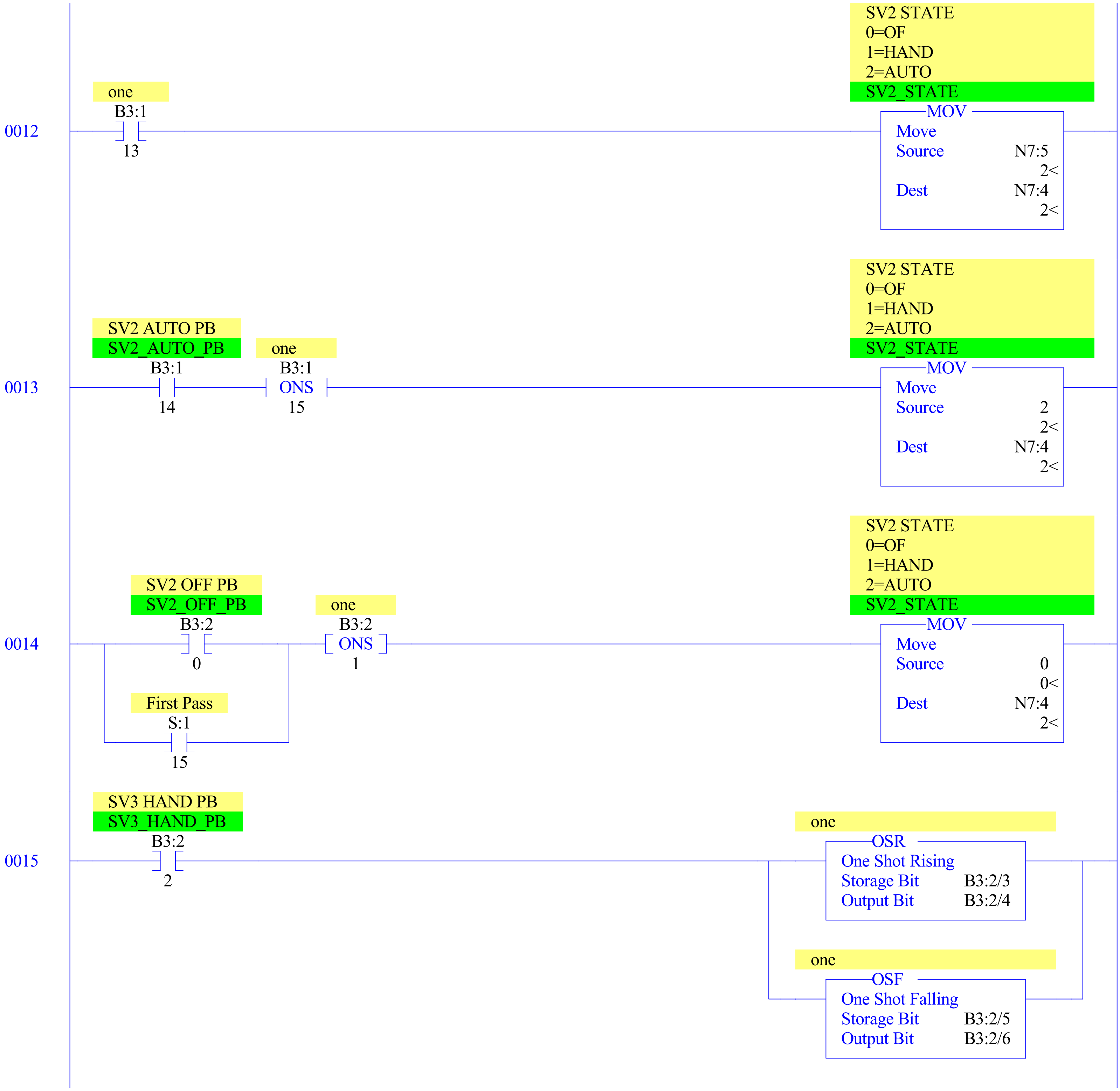
0012

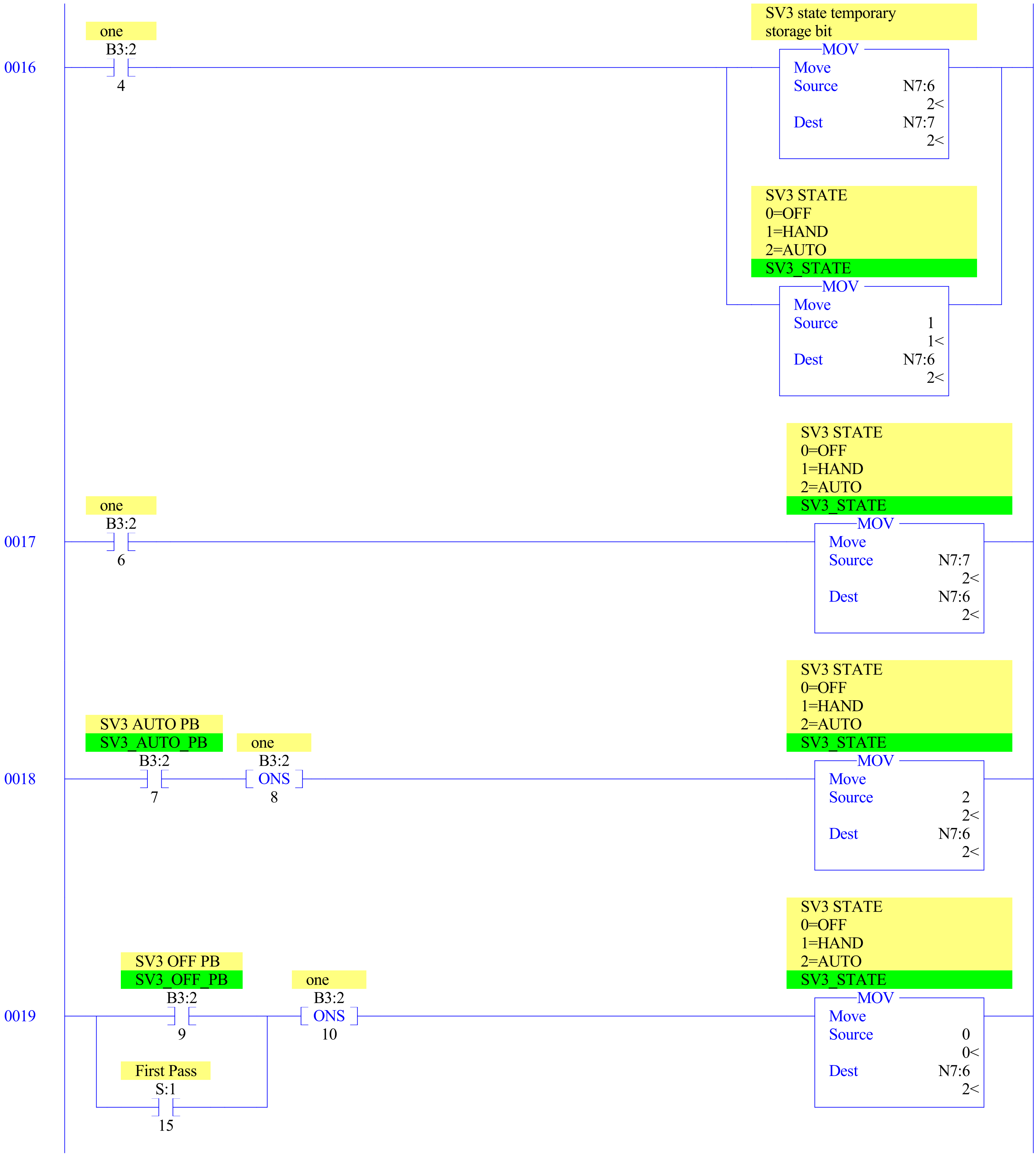
END


0000
P1 HAND PB
P1_HAND_PB
B3:0
8
one
OSR
One Shot Rising
Storage Bit B3:0/9
Output Bit B3:0/10
one
OSF
One Shot Falling
Storage Bit B3:0/11
Output Bit B3:0/12
0001
one
B3:0
10
P1 state temporary storage
MOV
Move
Source N7:0
2<
Dest N7:1
2<
P1 STATE
0=OFF
1=HAND
2=AUTO
P1_STATE
MOV
Move
Source 1
1<
Dest N7:0
2<
0002
one
B3:0
12
P1 STATE
0=OFF
1=HAND
2=AUTO
P1_STATE
MOV
Move
Source N7:1
2<
Dest N7:0
2<
0003
P1 AUTO PB
P1_AUTO_PB
B3:0
13
one
B3:0
ONS
14
P1 STATE
0=OFF
1=HAND
2=AUTO
P1_STATE
MOV
Move
Source 2
2<
Dest N7:0
2<


0004
P1 OFF PB
P1_OFF_PB
B3:0
15
First Pass
S:1
15
one
B3:1
ONS
0
P1 STATE
0=OFF
1=HAND
2=AUTO
P1_STATE
MOV
Move
Source 0
0<
Dest N7:0
2<
0005
SV1 HAND PB
SV1_HAND_PB
B3:1
1
one
OSR
One Shot Rising
Storage Bit B3:1/2
Output Bit B3:1/3
one
OSF
One Shot Falling
Storage Bit B3:1/4
Output Bit B3:1/5
0006
one
B3:1
3
SV1 temporary
storage bit
MOV
Move
Source N7:2
2<
Dest N7:3
2<
SV1 STATE
0=OFF
1=HAND
2=AUTO
SV1_STATE
MOV
Move
Source 1
1<
Dest N7:2
2<
0007
one
B3:1
5
SV1 STATE
0=OFF
1=HAND
2=AUTO
SV1_STATE
MOV
Move
Source N7:3
2<
Dest N7:2
2<


0008
SV1 AUTO PB
SV1_AUTO_PB
B3:1
6
one
B3:1
ONS
7
SV1 STATE
0=OFF
1=HAND
2=AUTO
SV1_STATE
MOV
Move
Source 2
2<
Dest N7:2
2<
0009
SV1 OFF PB
SV1_OFF_PB
B3:1
8
First Pass
S:1
15
one
B3:1
ONS
0
SV1 STATE
0=OFF
1=HAND
2=AUTO
SV1_STATE
MOV
Move
Source 0
0<
Dest N7:2
2<
0010
SV2 HAND PB
SV2_HAND_PB
B3:1
9
one
OSR
One Shot Rising
Storage Bit B3:1/10
Output Bit B3:1/11
one
OSF
One Shot Falling
Storage Bit B3:1/12
Output Bit B3:1/13
0011
one
B3:1
11
SV2 temporary
storage bit
MOV
Move
Source N7:4
2<
Dest N7:5
2<
SV2 STATE
0=OF
1=HAND
2=AUTO
SV2_STATE
MOV
Move
Source 1
1<
Dest N7:4
2<


0012
one
B3:1
13
SV2 STATE
0=OF
1=HAND
2=AUTO
SV2_STATE
MOV
Move
Source N7:5
2<
Dest N7:4
2<
0013
SV2 AUTO PB
SV2_AUTO_PB
B3:1
14
one
B3:1
ONS
15
SV2 STATE
0=OF
1=HAND
2=AUTO
SV2_STATE
MOV
Move
Source 2
2<
Dest N7:4
2<
0014
SV2 OFF PB
SV2_OFF_PB
B3:2
0
First Pass
S:1
15
one
B3:2
ONS
1
SV2 STATE
0=OF
1=HAND
2=AUTO
SV2_STATE
MOV
Move
Source 0
0<
Dest N7:4
2<
0015
SV3 HAND PB
SV3_HAND_PB
B3:2
2
one
OSR
One Shot Rising
Storage Bit B3:2/3
Output Bit B3:2/4
one
OSF
One Shot Falling
Storage Bit B3:2/5
Output Bit B3:2/6


0016
one
B3:2
4
SV3 state temporary storage bit
MOV
Move
Source N7:6
2<
Dest N7:7
2<
SV3 STATE
0=OFF
1=HAND
2=AUTO
SV3_STATE
MOV
Move
Source 1
1<
Dest N7:6
2<
0017
one
B3:2
6
SV3 STATE
0=OFF
1=HAND
2=AUTO
SV3_STATE
MOV
Move
Source N7:7
2<
Dest N7:6
2<
0018
SV3 AUTO PB
SV3_AUTO_PB
B3:2
7
one
B3:2
ONS
8
SV3 STATE
0=OFF
1=HAND
2=AUTO
SV3_STATE
MOV
Move
Source 2
2<
Dest N7:6
2<
0019
SV3 OFF PB
SV3_OFF_PB
B3:2
9
First Pass
S:1
15
one
B3:2
ONS
10
SV3 STATE
0=OFF
1=HAND
2=AUTO
SV3_STATE
MOV
Move
Source 0
0<
Dest N7:6
2<

0020
SV4 HAND PB
SV4_HAND_PB
B3:2
11

one
OSR
One Shot Rising
Storage Bit B3:2/12
Output Bit B3:2/13

one
OSF
One Shot Falling
Storage Bit B3:2/14
Output Bit B3:2/15

0021
one
B3:2
13

SV4 State temporary
storage bit
MOV
Move
Source N7:8
2<
Dest N7:9
0<

SV4 STATE
0=OFF
1=HAND
1=AUTO
SV4_STATE
MOV
Move
Source 1
1<
Dest N7:8
2<

0022
one
B3:2
15

SV4 STATE
0=OFF
1=HAND
1=AUTO
SV4_STATE
MOV
Move
Source N7:9
0<
Dest N7:8
2<

0023
SV4 AUTO PB
SV4_AUTO_PB
B3:3
0

one
B3:3
ONS
1

SV4 STATE
0=OFF
1=HAND
1=AUTO
SV4_STATE
MOV
Move
Source 2
2<
Dest N7:8
2<

0024

SV4 OFF PB
SV4_OFF_PB
B3:3
2

First Pass
S:1
15

one
B3:3
ONS
3

SV4 STATE
0=OFF
1=HAND
1=AUTO
SV4_STATE
MOV
Move
Source 0
0<
Dest N7:8
2<

0025

SV5 HAND PB
SV5_HAND_PB
B3:3
4

one
OSR
One Shot Rising
Storage Bit B3:3/5
Output Bit B3:3/6

one
OSF
One Shot Falling
Storage Bit B3:3/7
Output Bit B3:3/8

0026

one
B3:3
6

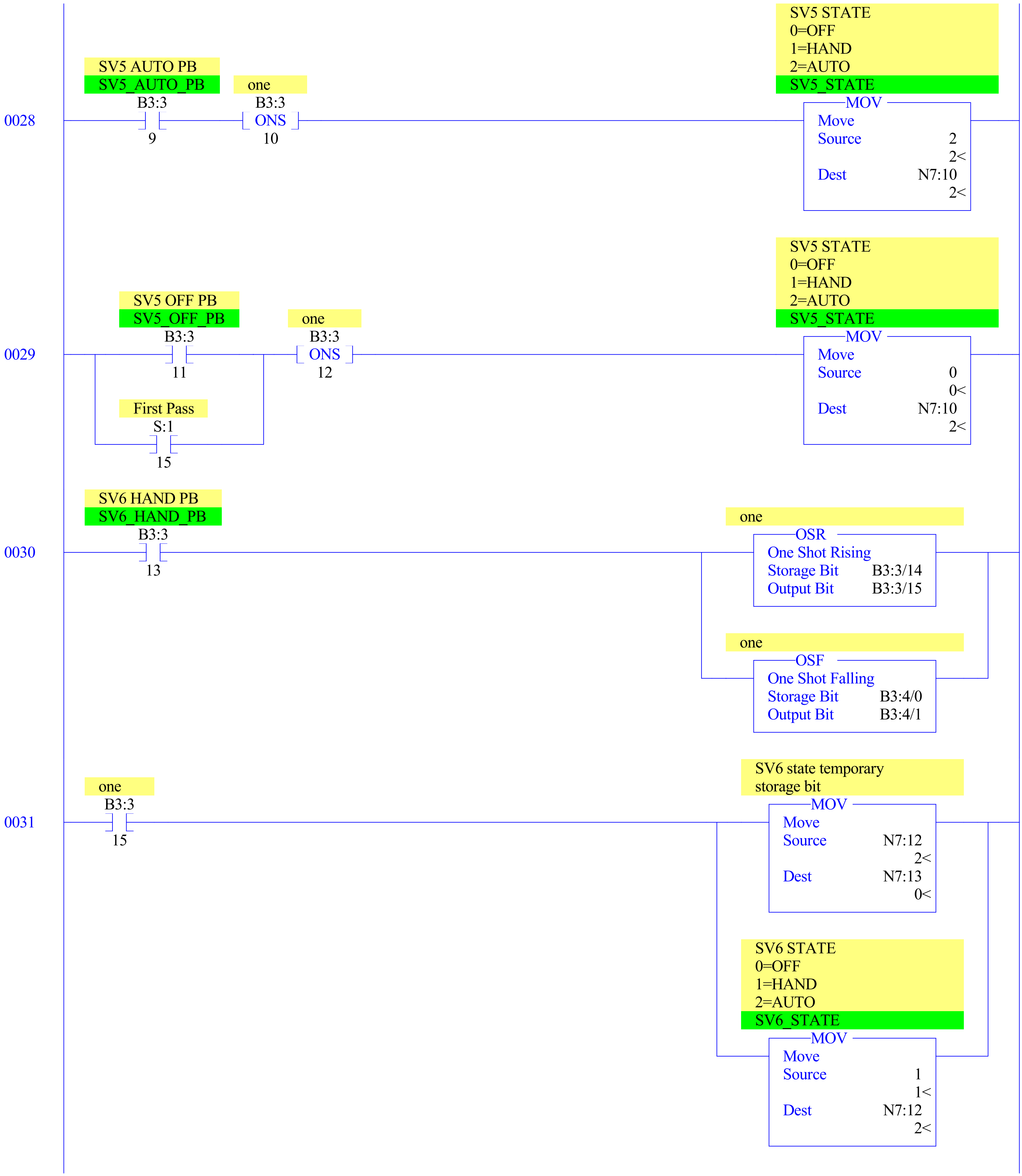
SV5 state temporary
storage bit
MOV
Move
Source N7:10
2<
Dest N7:11
2<

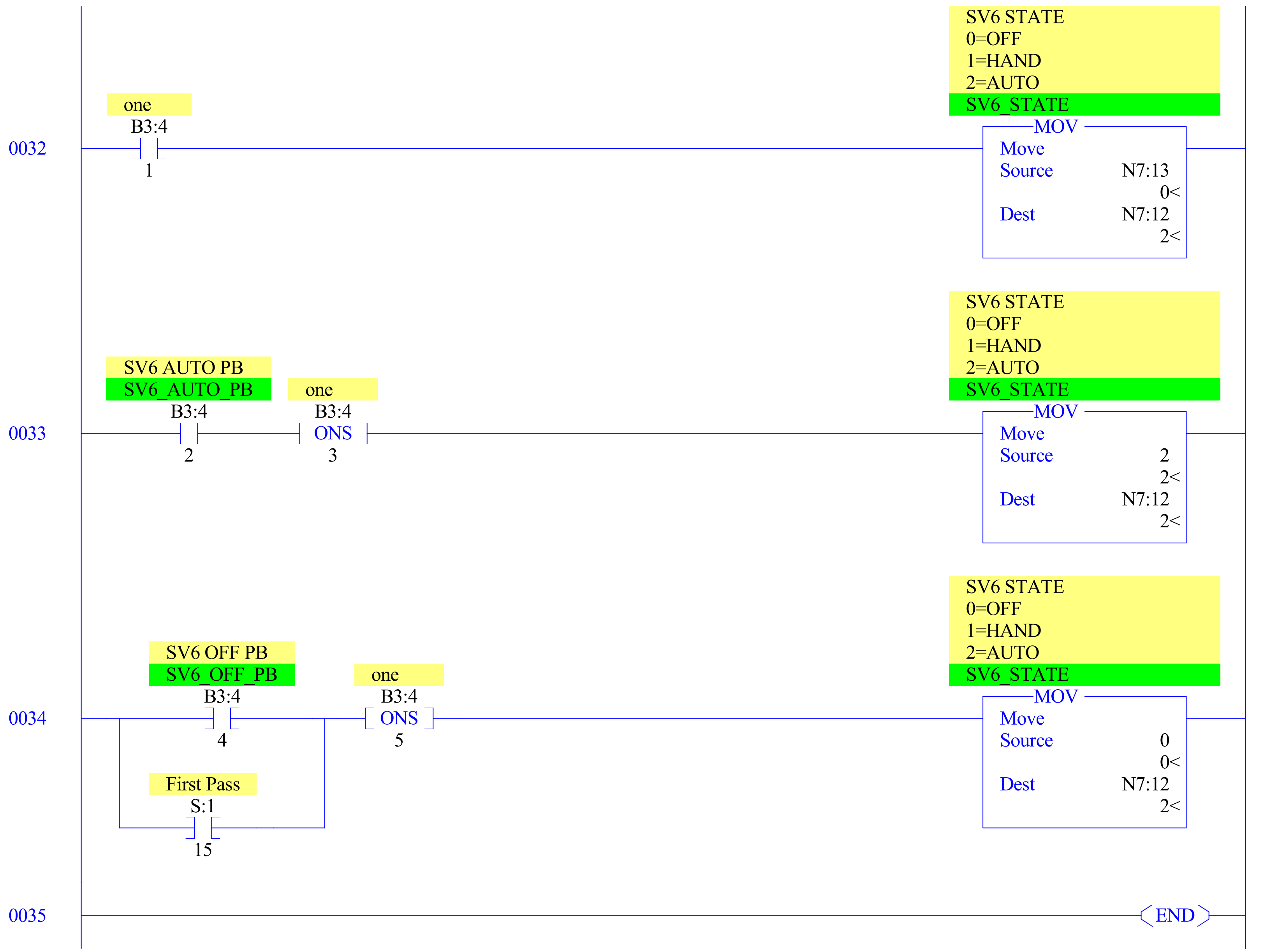
SV5 STATE
0=OFF
1=HAND
2=AUTO
SV5_STATE
MOV
Move
Source 1
1<
Dest N7:10
2<

0027

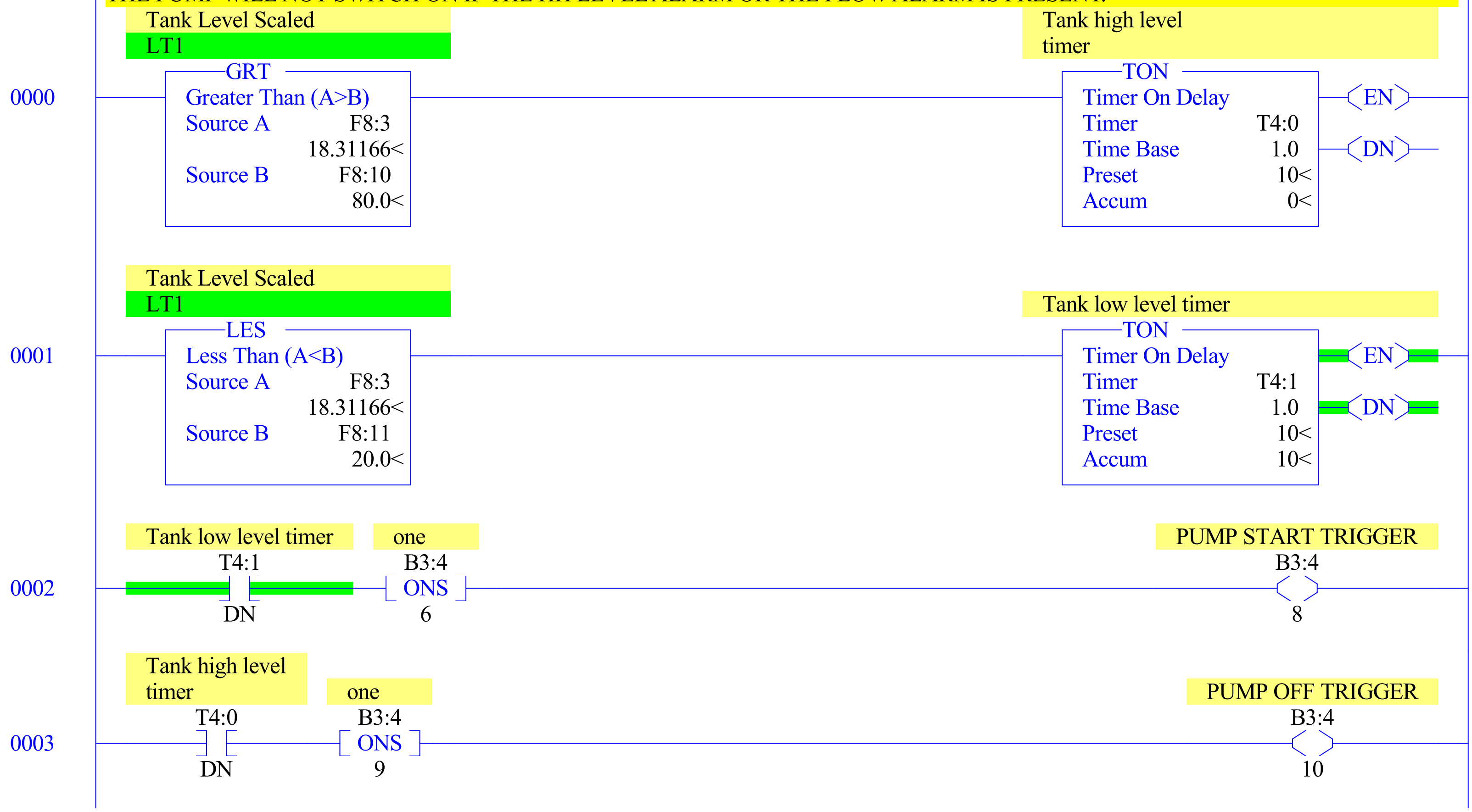
one
B3:3
8

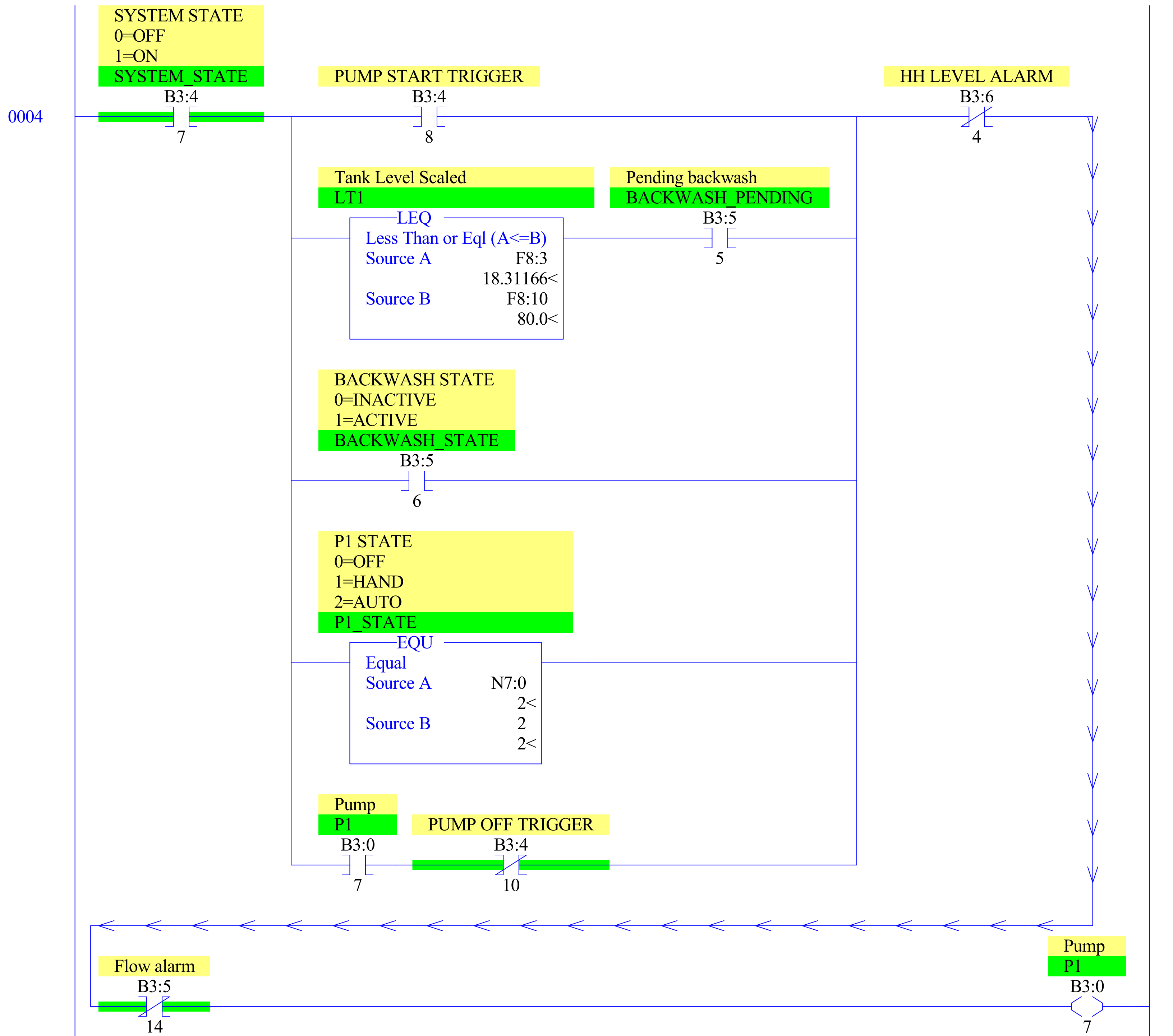
SV5 STATE
0=OFF
1=HAND
2=AUTO
SV5_STATE
MOV
Move
Source N7:11
2<
Dest N7:10
2<


0028
SV5 AUTO PB
SV5_AUTO_PB
B3:3
9
one
B3:3
ONS
10
SV5 STATE
0=OFF
1=HAND
2=AUTO
SV5_STATE
MOV
Move
Source 2
2<
Dest N7:10
2<
0029
SV5 OFF PB
SV5_OFF_PB
B3:3
11
First Pass
S:1
15
one
B3:3
ONS
12
SV5 STATE
0=OFF
1=HAND
2=AUTO
SV5_STATE
MOV
Move
Source 0
0<
Dest N7:10
2<
0030
SV6 HAND PB
SV6_HAND_PB
B3:3
13
one
OSR
One Shot Rising
Storage Bit B3:3/14
Output Bit B3:3/15
one
OSF
One Shot Falling
Storage Bit B3:4/0
Output Bit B3:4/1
0031
one
B3:3
15
SV6 state temporary
storage bit
MOV
Move
Source N7:12
2<
Dest N7:13
0<
SV6 STATE
0=OFF
1=HAND
2=AUTO
SV6_STATE
MOV
Move
Source 1
1<
Dest N7:12
2<

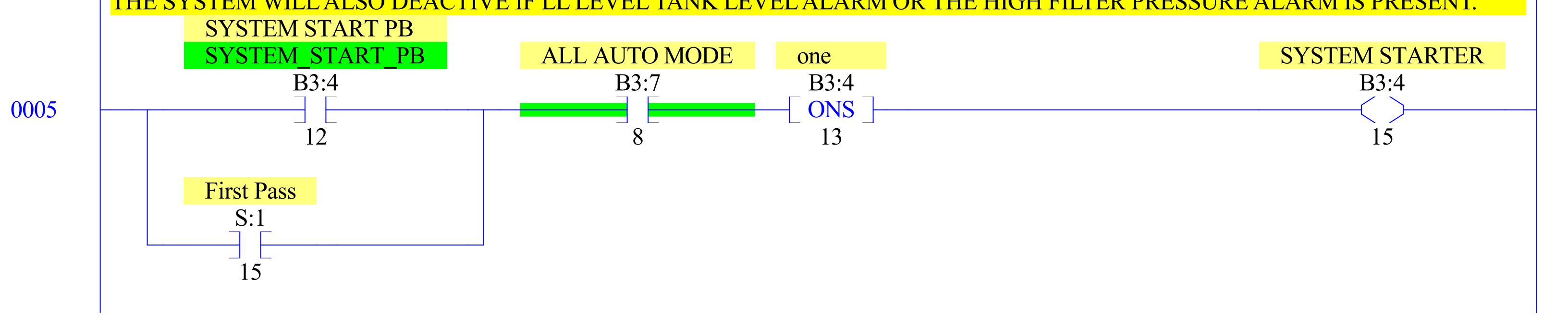

0032
one
B3:4
1
SV6 STATE
0=OFF
1=HAND
2=AUTO
SV6_STATE
MOV
Move
Source N7:13
0<
Dest N7:12
2<
0033
SV6 AUTO PB
SV6_AUTO_PB
B3:4
2
one
B3:4
ONS
3
SV6 STATE
0=OFF
1=HAND
2=AUTO
SV6_STATE
MOV
Move
Source 2
2<
Dest N7:12
2<
0034
SV6 OFF PB
SV6_OFF_PB
B3:4
4
First Pass
S:1
15
one
B3:4
ONS
5
SV6 STATE
0=OFF
1=HAND
2=AUTO
SV6_STATE
MOV
Move
Source 0
0<
Dest N7:12
2<
0035
END

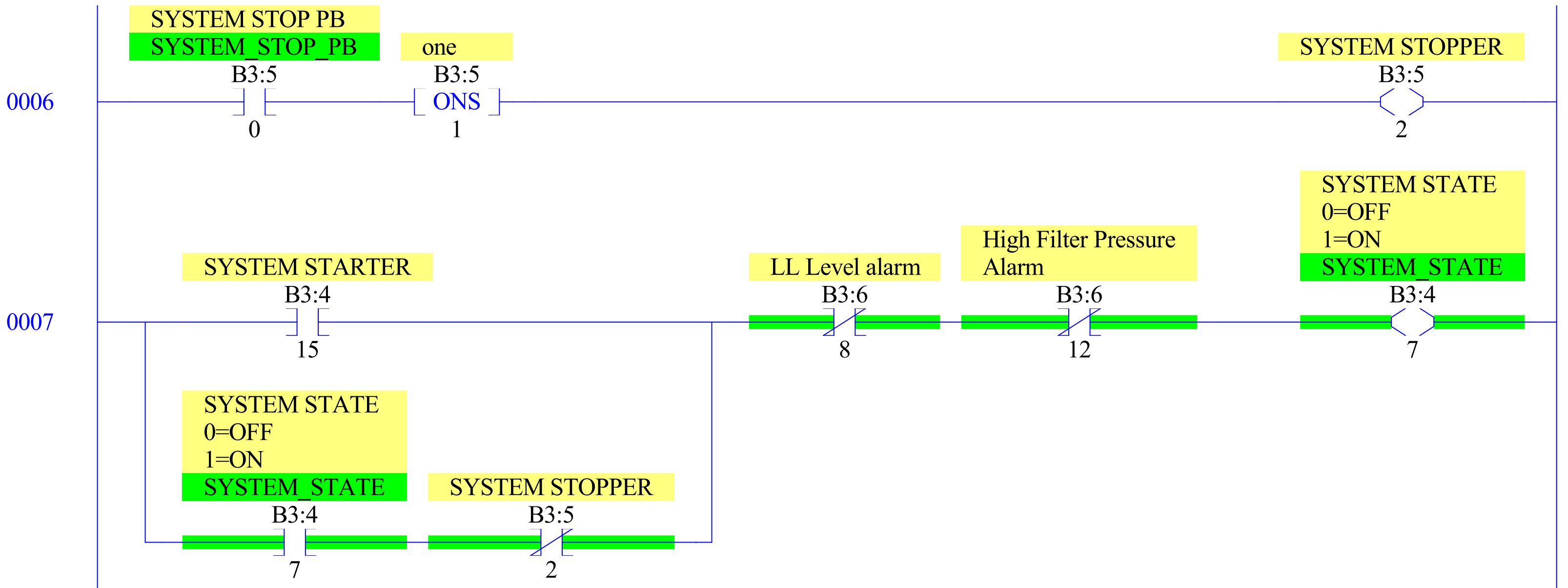
PUMP CONTROL:
THE PUMP IS ON AS LONG AS THE HIGH TANK LEVEL SPECIFIED IS REACHED. WHEN THAT LEVEL IS REACHED, THE PUMP WILL SWITCH OFF, AND THE DRAINING PROCESS BEGINS.
WHEN THE TANK IS DRAINING, THE PUMP WILL REMAIN SWITCHED OFF UNTIL THE SPECIFIED TANK LOW LEVEL IS REACHED.
THE PUMP WILL NOT SWITCH ON IF THE HH LEVEL ALARM OR THE FLOW ALARM IS PRESENT.



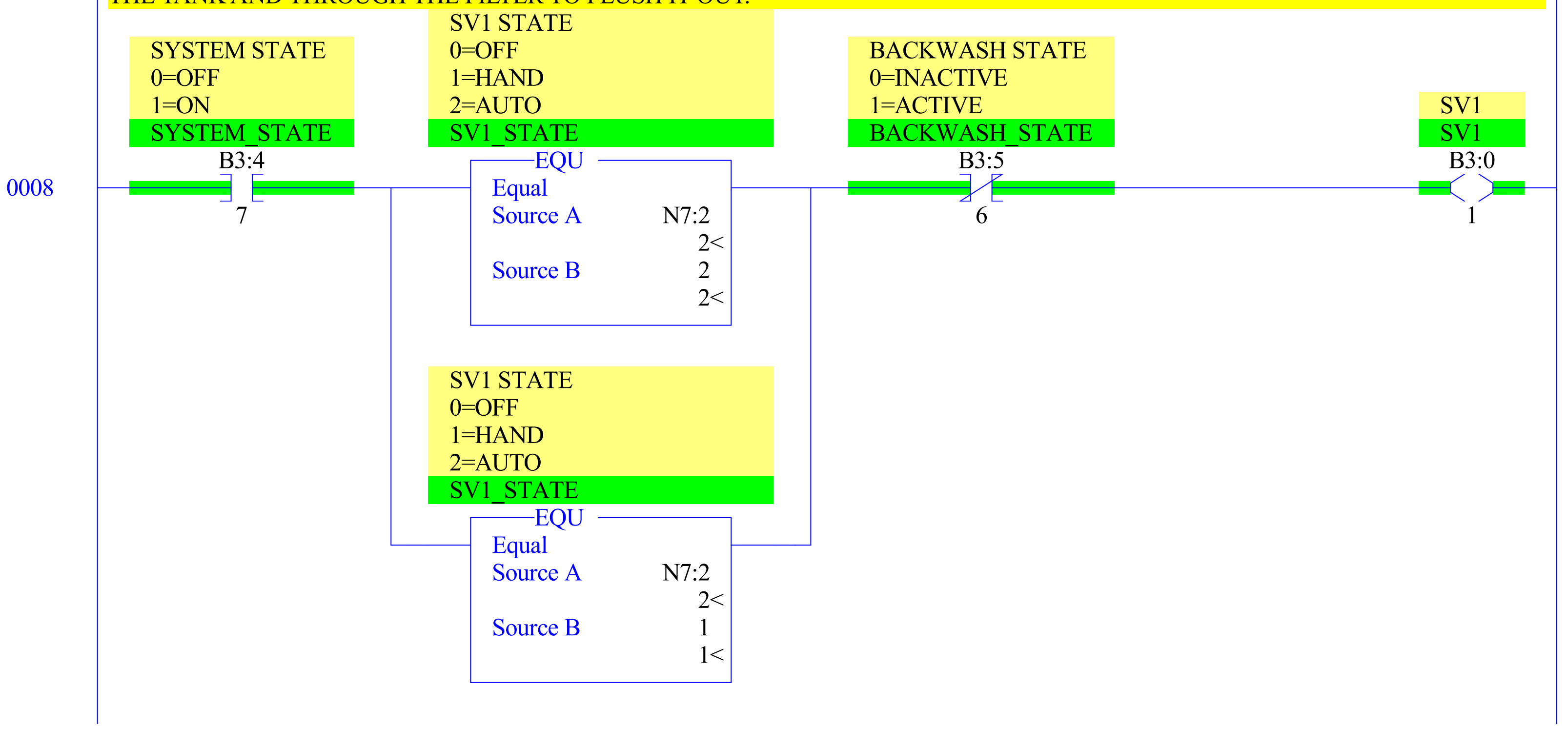


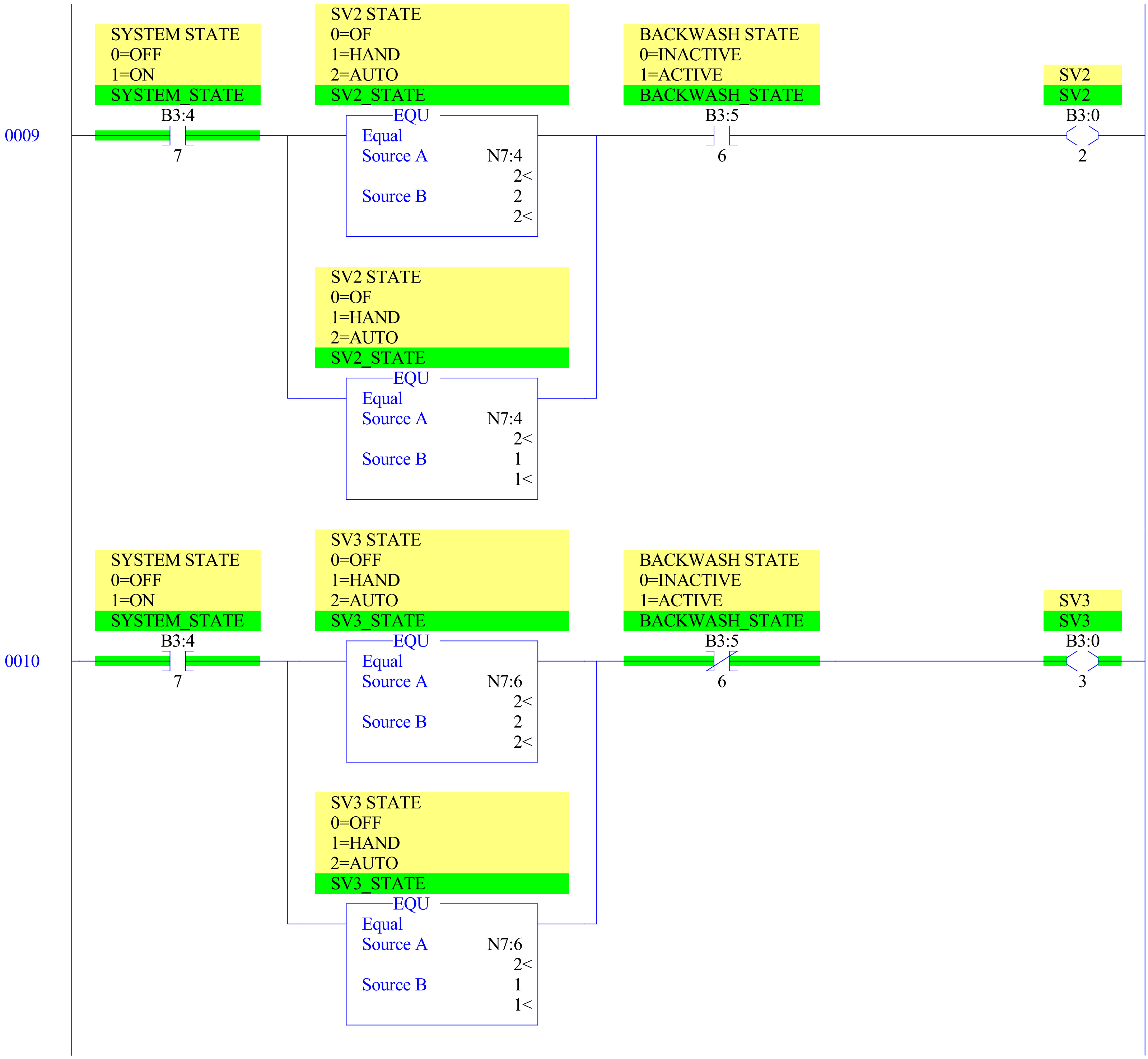
SYSTEM MODE CONTROL:
THE SYSTEM IS ACTIVATED WHEN ALL VALVES AND THE PUMP ARE IN AUTO MODE AND WHEN THE SYSTEM START PUSHBUTTON IS PRESSED.
THE SYSTEM WILL DEACTIVATE AGAIN WHEN THE STOP PUSHBUTTON IS PRESSED.
THE SYSTEM WILL ALSO DEACTIVE IF LL LEVEL TANK LEVEL ALARM OR THE HIGH FILTER PRESSURE ALARM IS PRESENT.

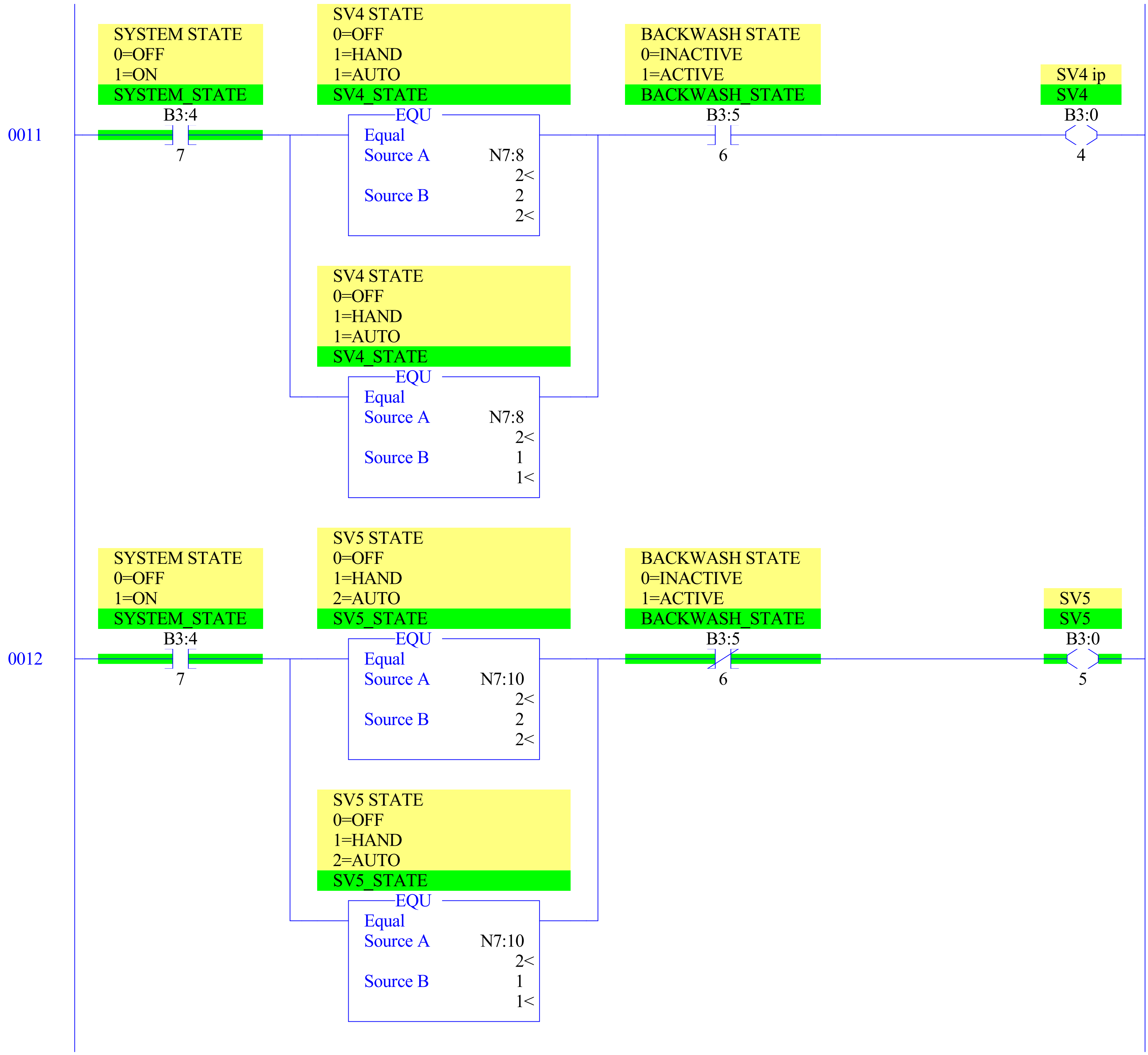


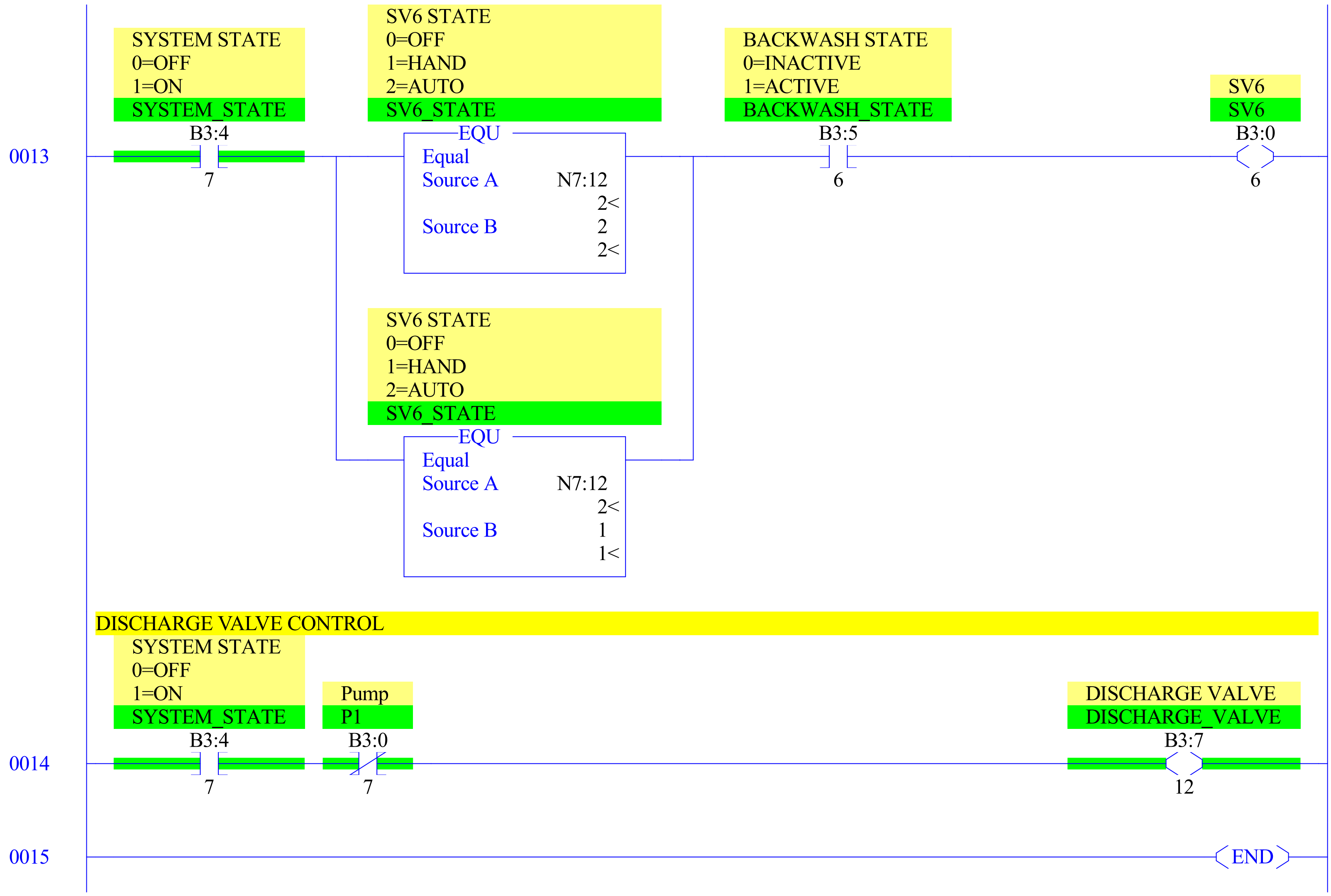


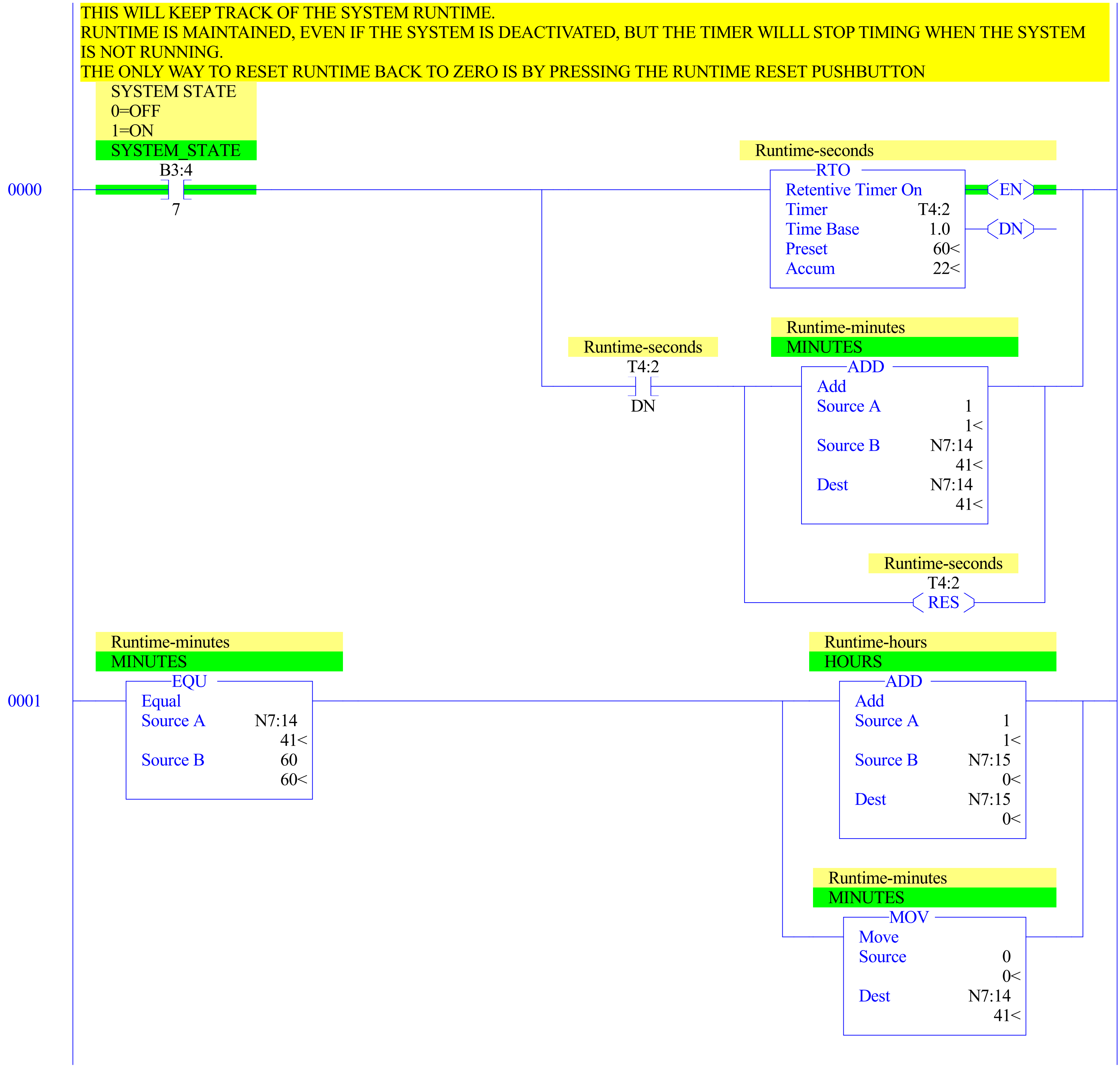
VALVE CONTROL:
VALVES 1, 3, AND 5 ARE ENERGIZED WHEN BACKWASH IS INACTIVE. THESE VALVES WILL DEENERGIZE WHEN THE BACKWASH CYCLE IS ACTIVE.
VALVES 2, 4, AND 6 ARE ENERGIZED WHEN THE BACKWASH CYCLE IS ACTIVE, ALLOWING FOR THE FLOW OF WATER OUT OF THE TANK AND THROUGH THE FILTER TO FLUSH IT OUT.

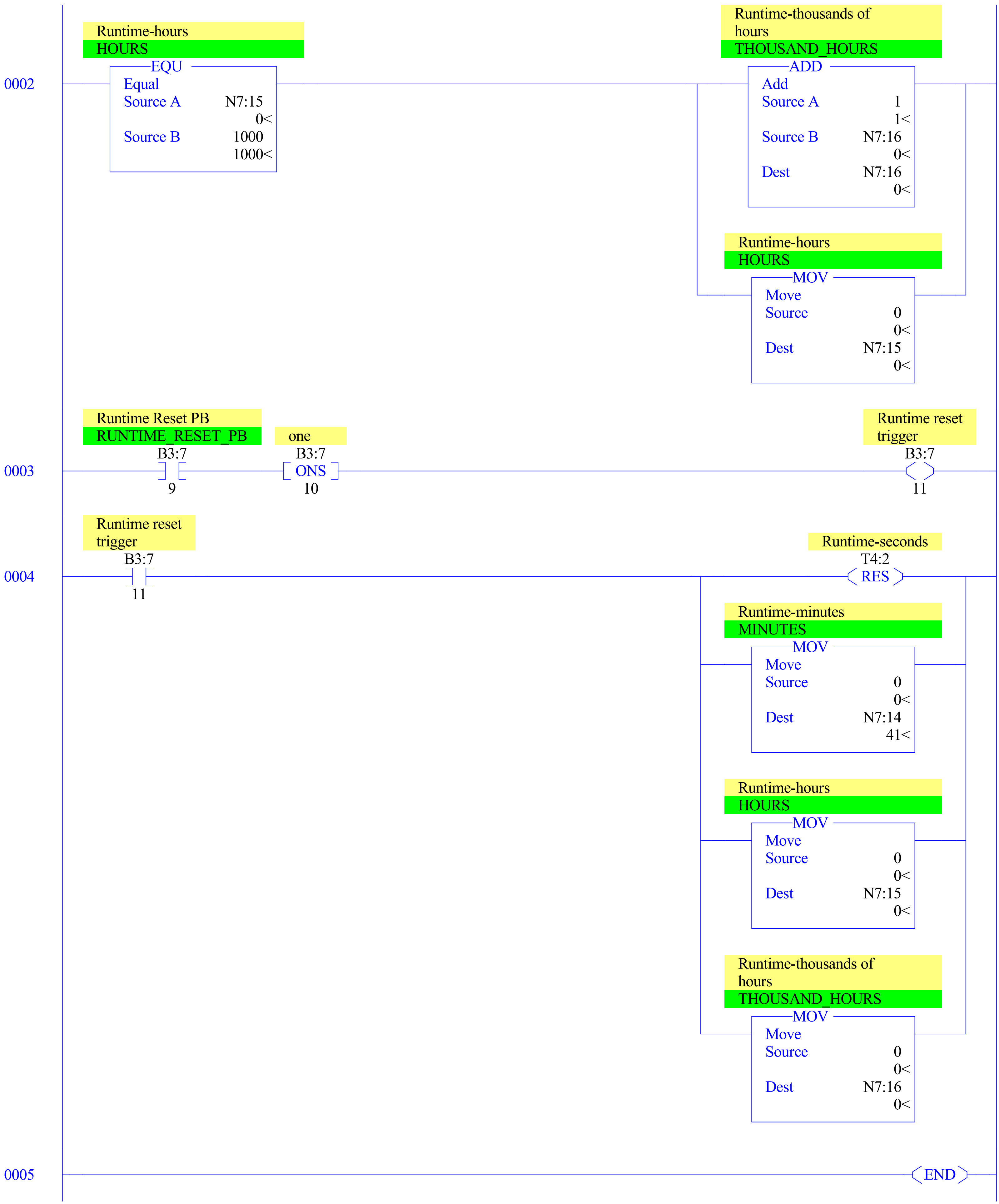



0009
SYSTEM STATE
0=OFF
1=ON
SYSTEM_STATE
B3:4
7
SV2 STATE
0=OF
1=HAND
2=AUTO
SV2_STATE
EQU
Equal
Source A N7:4
2<
Source B 2
2<
SV2 STATE
0=OF
1=HAND
2=AUTO
SV2_STATE
EQU
Equal
Source A N7:4
2<
Source B 1
1<
BACKWASH STATE
0=INACTIVE
1=ACTIVE
BACKWASH_STATE
B3:5
6
SV2
SV2
B3:0
2
0010
SYSTEM STATE
0=OFF
1=ON
SYSTEM_STATE
B3:4
7
SV3 STATE
0=OFF
1=HAND
2=AUTO
SV3_STATE
EQU
Equal
Source A N7:6
2<
Source B 2
2<
SV3 STATE
0=OFF
1=HAND
2=AUTO
SV3_STATE
EQU
Equal
Source A N7:6
2<
Source B 1
1<
BACKWASH STATE
0=INACTIVE
1=ACTIVE
BACKWASH_STATE
B3:5
6
SV3
SV3
B3:0
3


0011
SYSTEM STATE
0=OFF
1=ON
SYSTEM_STATE
B3:4
7
SV4 STATE
0=OFF
1=HAND
1=AUTO
SV4_STATE
EQU
Equal
Source A N7:8
2<
Source B 2
2<
SV4 STATE
0=OFF
1=HAND
1=AUTO
SV4_STATE
EQU
Equal
Source A N7:8
2<
Source B 1
1<
BACKWASH STATE
0=INACTIVE
1=ACTIVE
BACKWASH_STATE
B3:5
6
SV4 ip
SV4
B3:0
4
0012
SYSTEM STATE
0=OFF
1=ON
SYSTEM_STATE
B3:4
7
SV5 STATE
0=OFF
1=HAND
2=AUTO
SV5_STATE
EQU
Equal
Source A N7:10
2<
Source B 2
2<
SV5 STATE
0=OFF
1=HAND
2=AUTO
SV5_STATE
EQU
Equal
Source A N7:10
2<
Source B 1
1<
BACKWASH STATE
0=INACTIVE
1=ACTIVE
BACKWASH_STATE
B3:5
6
SV5
SV5
B3:0
5


0013
SYSTEM STATE
0=OFF
1=ON
SYSTEM_STATE
B3:4
7
SV6 STATE
0=OFF
1=HAND
2=AUTO
SV6_STATE
EQU
Equal
Source A N7:12
2<
Source B 2
2<
BACKWASH STATE
0=INACTIVE
1=ACTIVE
BACKWASH_STATE
B3:5
6
SV6
SV6
B3:0
6
SV6 STATE
0=OFF
1=HAND
2=AUTO
SV6_STATE
EQU
Equal
Source A N7:12
2<
Source B 1
1<
DISCHARGE VALVE CONTROL
0014
SYSTEM STATE
0=OFF
1=ON
SYSTEM_STATE
B3:4
7
Pump
P1
B3:0
7
DISCHARGE VALVE
DISCHARGE_VALVE
B3:7
12
0015
END


THIS WILL KEEP TRACK OF THE SYSTEM RUNTIME.
RUNTIME IS MAINTAINED, EVEN IF THE SYSTEM IS DEACTIVATED, BUT THE TIMER WILLL STOP TIMING WHEN THE SYSTEM IS NOT RUNNING.
THE ONLY WAY TO RESET RUNTIME BACK TO ZERO IS BY PRESSING THE RUNTIME RESET PUSHBUTTON
0000
SYSTEM STATE
0=OFF
1=ON
SYSTEM_STATE
B3:4
7
Runtime-seconds
RTO
Retentive Timer On
Timer T4:2
Time Base 1.0
Preset 60<
Accum 22<
EN
DN
Runtime-seconds
T4:2
DN
Runtime-minutes
MINUTES
ADD
Add
Source A 1
1<
Source B N7:14
41<
Dest N7:14
41<
Runtime-seconds
T4:2
RES
0001
Runtime-minutes
MINUTES
EQU
Equal
Source A N7:14
41<
Source B 60
60<
Runtime-hours
HOURS
ADD
Add
Source A 1
1<
Source B N7:15
0<
Dest N7:15
0<
Runtime-minutes
MINUTES
MOV
Move
Source 0
0<
Dest N7:14
41<


0002
Runtime-hours
HOURS
EQU
Equal
Source A N7:15
0<
Source B 1000
1000<
Runtime-thousands of hours
THOUSAND_HOURS
ADD
Add
Source A 1
1<
Source B N7:16
0<
Dest N7:16
0<
Runtime-hours
HOURS
MOV
Move
Source 0
0<
Dest N7:15
0<
0003
Runtime Reset PB
RUNTIME_RESET_PB
B3:7
9
one
B3:7
ONS
10
Runtime reset trigger
B3:7
11
0004
Runtime reset trigger
B3:7
11
Runtime-seconds
T4:2
RES
Runtime-minutes
MINUTES
MOV
Move
Source 0
0<
Dest N7:14
41<
Runtime-hours
HOURS
MOV
Move
Source 0
0<
Dest N7:15
0<
Runtime-thousands of hours
THOUSAND_HOURS
MOV
Move
Source 0
0<
Dest N7:16
0<
0005
END

CALCULATING THE DIFFERENTIAL PRESSURE IN THE FILTER

0000

Diff. pressure
SUB
Subtract
Source A F8:0
0.06103888<
Source B F8:1
2.288958<
Dest F8:4
-2.227919<

0001

Diff Pressure-
absolute
DIFF_PRESS_ABS
ABS
Absolute Value
Source F8:4
-2.227919<
Dest F8:5
2.227919<

ACTIVATING BACKWASH WHEN THE SPECIFIED FILTER DIFFERENTIAL PRESSURE IS REACHED.
PRESSING THE MANUAL BACKWASH PUSBUTTON WILL ALSO ACTIVATE A BACKWASH CYCLE.

0002

SYSTEM STATE
0=OFF
1=ON
SYSTEM_STATE
B3:4
7

Diff Pressure-
absolute
DIFF_PRESS_ABS
GRT
Greater Than (A>B)
Source A F8:5
2.227919<
Source B F8:6
12.0<

MANUAL BACKWASH
PUSHBUTTON
MANUAL_BW_PB
B3:7
6

one
B3:5
ONS
4

Pending backwash
trigger
B3:5
3

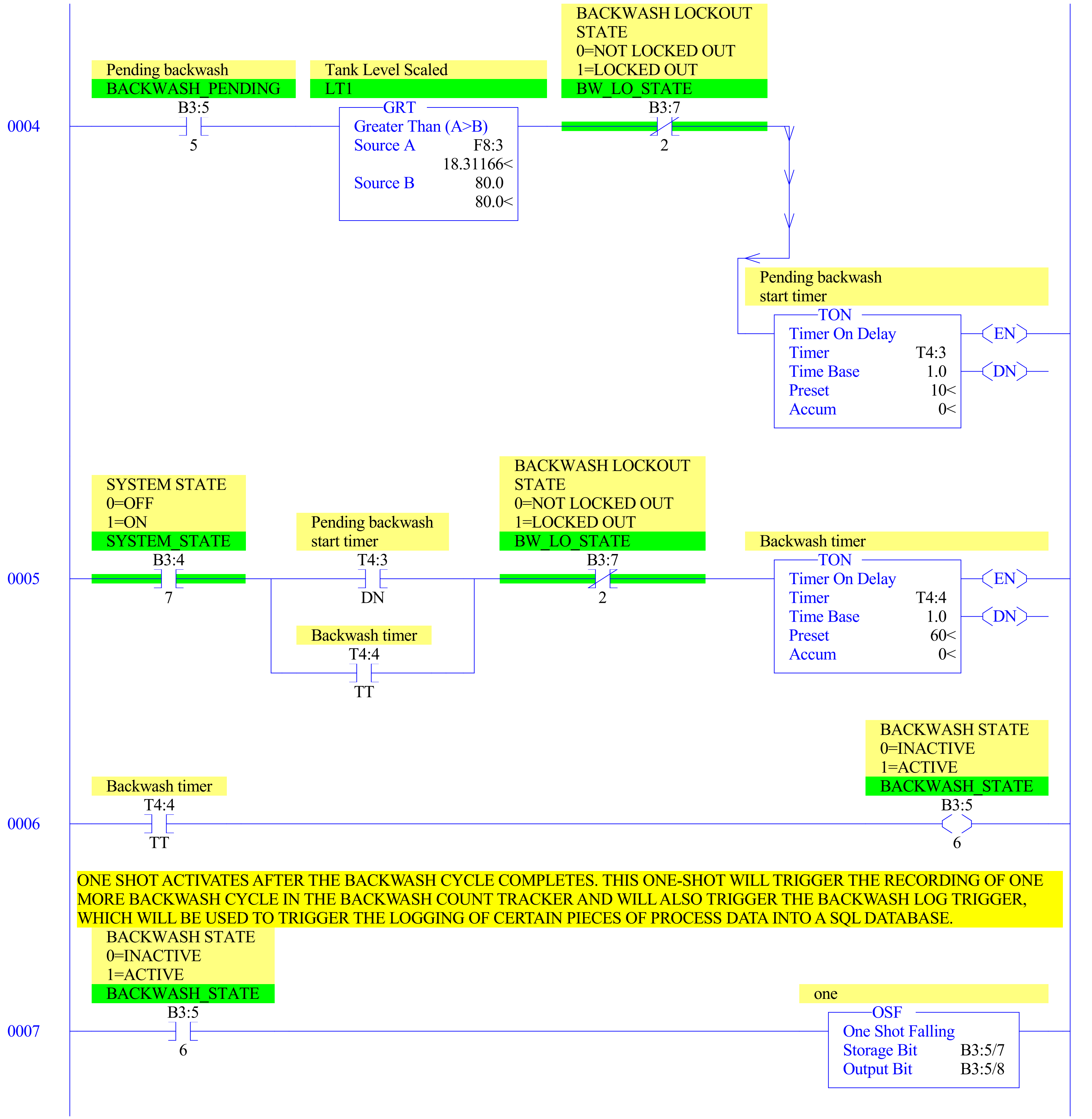
0003

Pending backwash
trigger
B3:5
3

Pending backwash
BACKWASH_PENDING
B3:5
5

Pending backwash
start timer
T4:3
DN

Pending backwash
BACKWASH_PENDING
B3:5
5


0004
Pending backwash
BACKWASH_PENDING
B3:5
5
Tank Level Scaled
LT1
GRT
Greater Than (A>B)
Source A F8:3
18.31166<
Source B 80.0
80.0<
BACKWASH LOCKOUT STATE
0=NOT LOCKED OUT
1=LOCKED OUT
BW_LO_STATE
B3:7
2
Pending backwash start timer
TON
Timer On Delay
Timer T4:3
Time Base 1.0
Preset 10<
Accum 0<
EN
DN
0005
SYSTEM STATE
0=OFF
1=ON
SYSTEM_STATE
B3:4
7
Pending backwash start timer
T4:3
DN
Backwash timer
T4:4
TT
BACKWASH LOCKOUT STATE
0=NOT LOCKED OUT
1=LOCKED OUT
BW_LO_STATE
B3:7
2
Backwash timer
TON
Timer On Delay
Timer T4:4
Time Base 1.0
Preset 60<
Accum 0<
EN
DN
0006
Backwash timer
T4:4
TT
BACKWASH STATE
0=INACTIVE
1=ACTIVE
BACKWASH_STATE
B3:5
6
ONE SHOT ACTIVATES AFTER THE BACKWASH CYCLE COMPLETES. THIS ONE-SHOT WILL TRIGGER THE RECORDING OF ONE MORE BACKWASH CYCLE IN THE BACKWASH COUNT TRACKER AND WILL ALSO TRIGGER THE BACKWASH LOG TRIGGER, WHICH WILL BE USED TO TRIGGER THE LOGGING OF CERTAIN PIECES OF PROCESS DATA INTO A SQL DATABASE.
0007
BACKWASH STATE
0=INACTIVE
1=ACTIVE
BACKWASH_STATE
B3:5
6
one
OSF
One Shot Falling
Storage Bit B3:5/7
Output Bit B3:5/8

0008

one
B3:5
8

Backwash Cycle Counter
BACKWASH_COUNT
ADD
Add
Source A 1
1<
Source B N7:17
0<
Dest N7:17
0<

Backwash End Data Log Trigger
BW_LOG_TRIGGER
B3:5
11

WHEN PRESSED FOR 5 SECONDS, THE BACKWASH COUNT RESET PUSHBUTTON WILL RESET BACKWASH COUNT.

0009

Backwash Counht rest PB
BW_COUNT_RESET_PB
B3:5
9

BW COUNT RESET PB TIMER
TON
Timer On Delay
Timer T4:5
Time Base 1.0
Preset 5<
Accum 0<
EN
DN

0010

BW COUNT RESET PB TIMER
T4:5
DN

one
B3:5
ONS
10

Backwash Cycle Counter
BACKWASH_COUNT
MOV
Move
Source 0
0<
Dest N7:17
0<

0011

Backwash time left
BW_TIME_LEFT
SUB
Subtract
Source A 60
60<
Source B T4:4.ACC
0<
Dest N7:18
60<

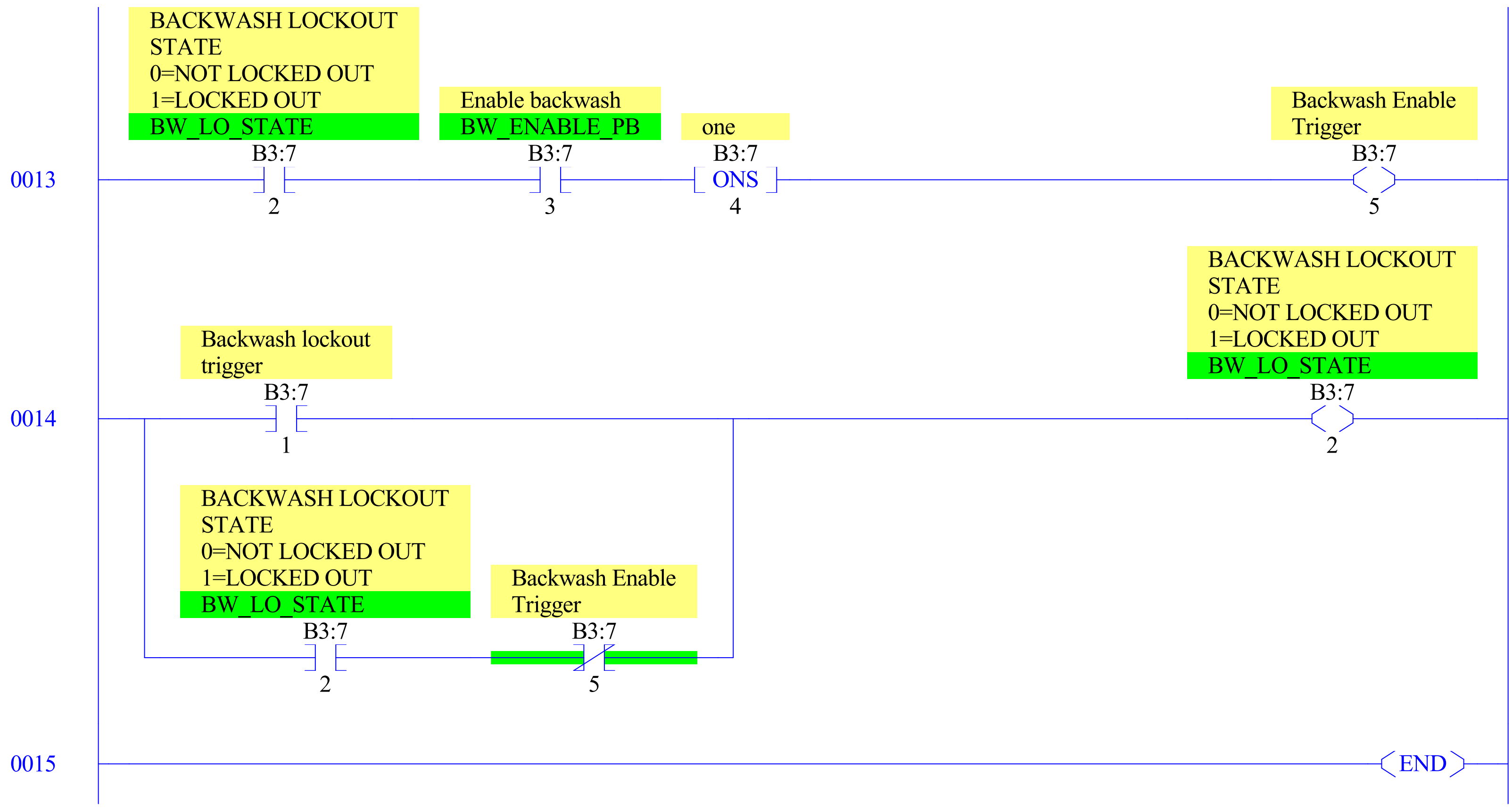
THE BACKWASH LOCKOUT PUSHBUTTON WILL LOCK OUT (DISABLE) THE BACKWASH CYCLE..
TO ENABLE THE BACKWASH CYCLE AGAIN, THE BACKWASH ENABLE PUSHBUTTON WILL HAVE TO BE PUSHED.

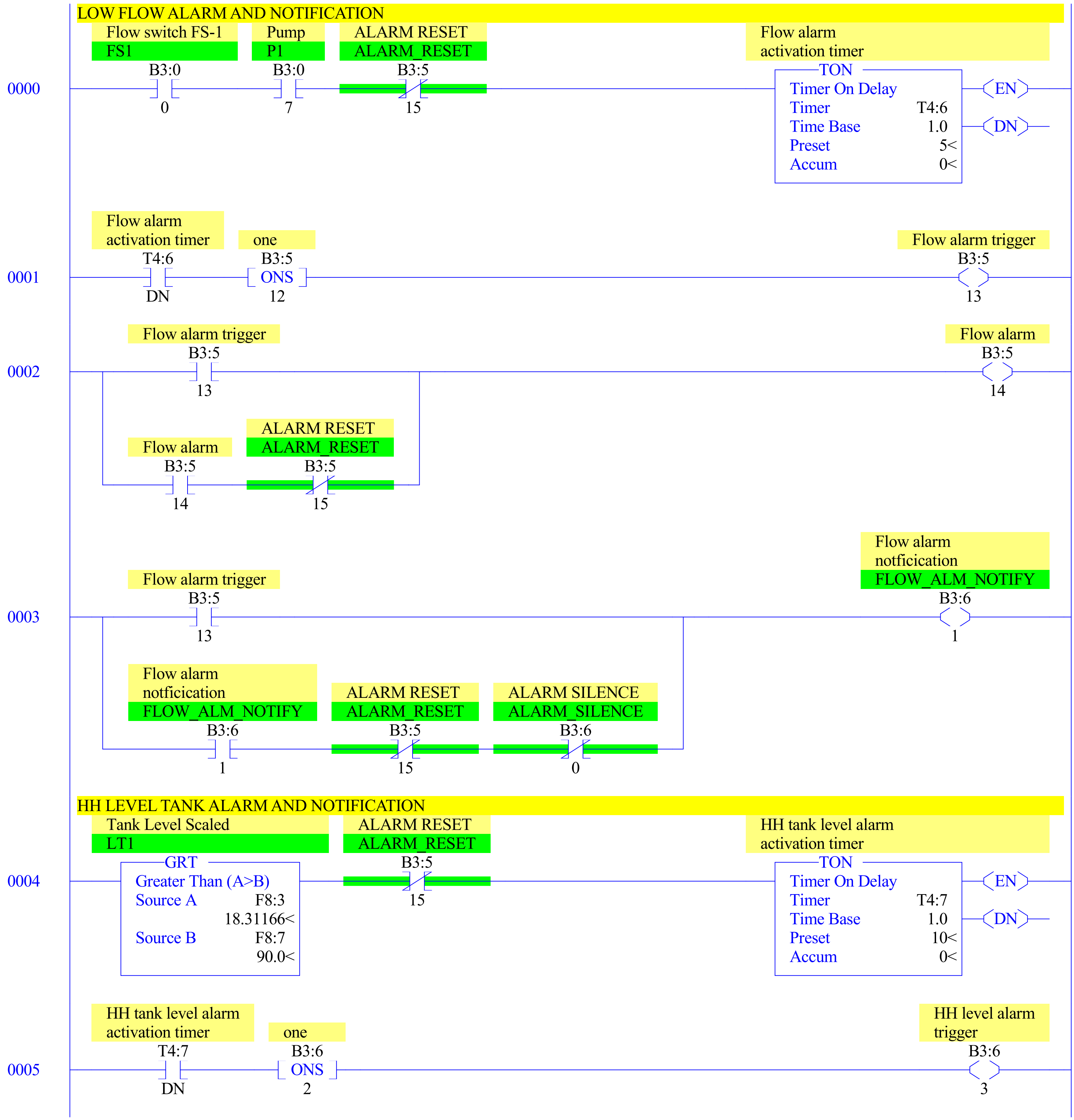
0012

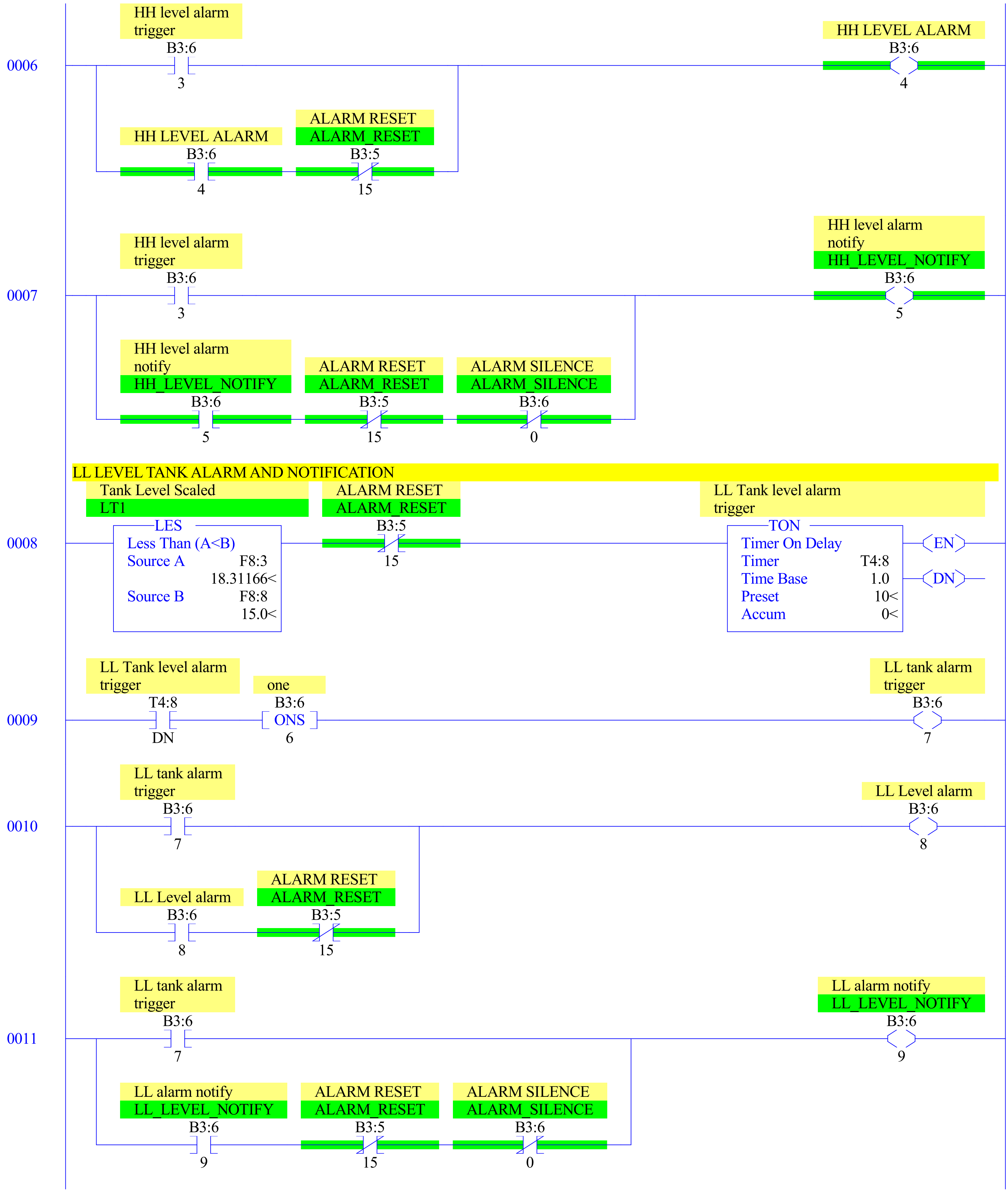
Lockout Backwash PB
BW_LOCKOUT_PB
B3:6
14

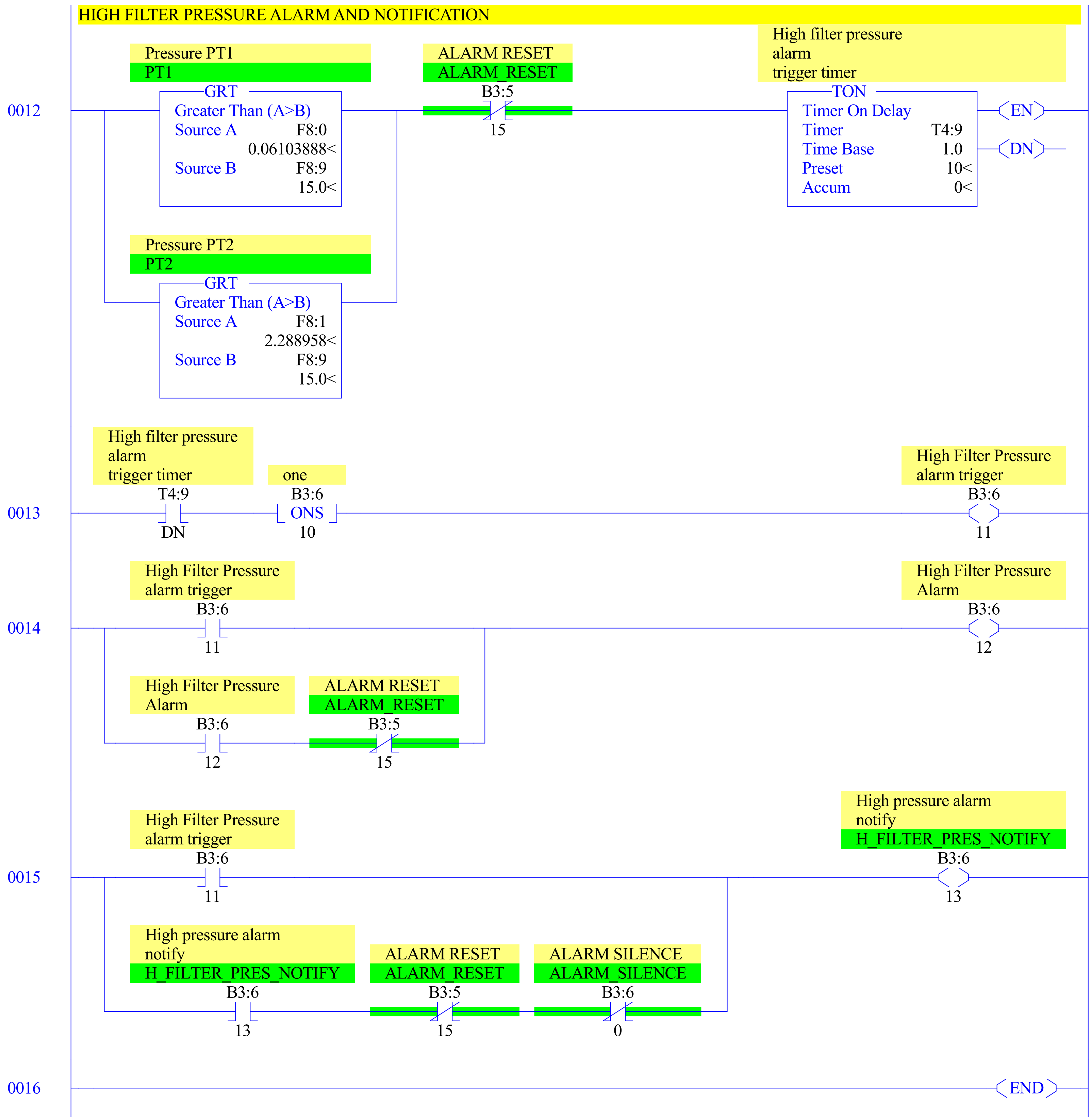
one
B3:7
ONS
0

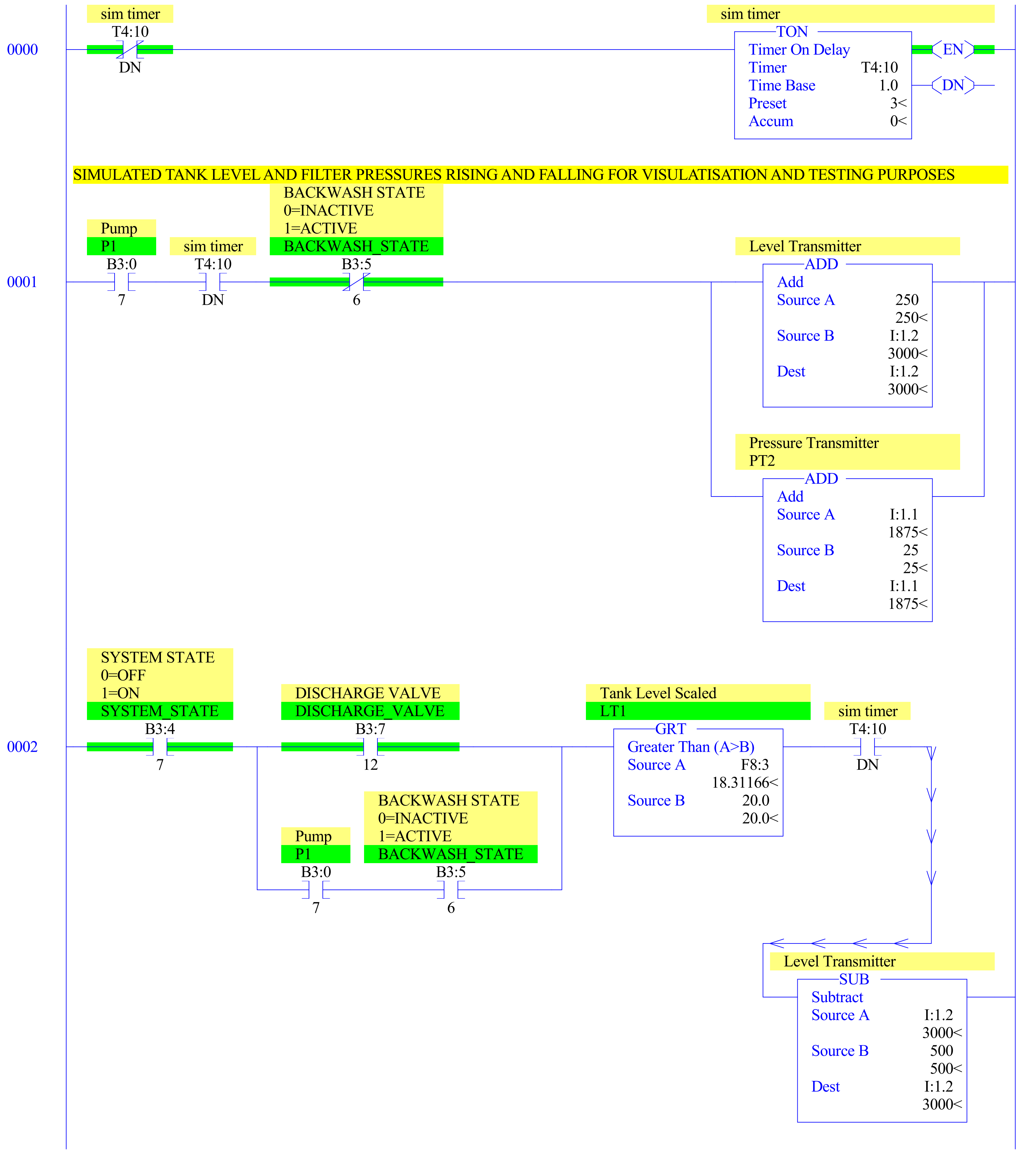
Backwash lockout trigger
B3:7
1

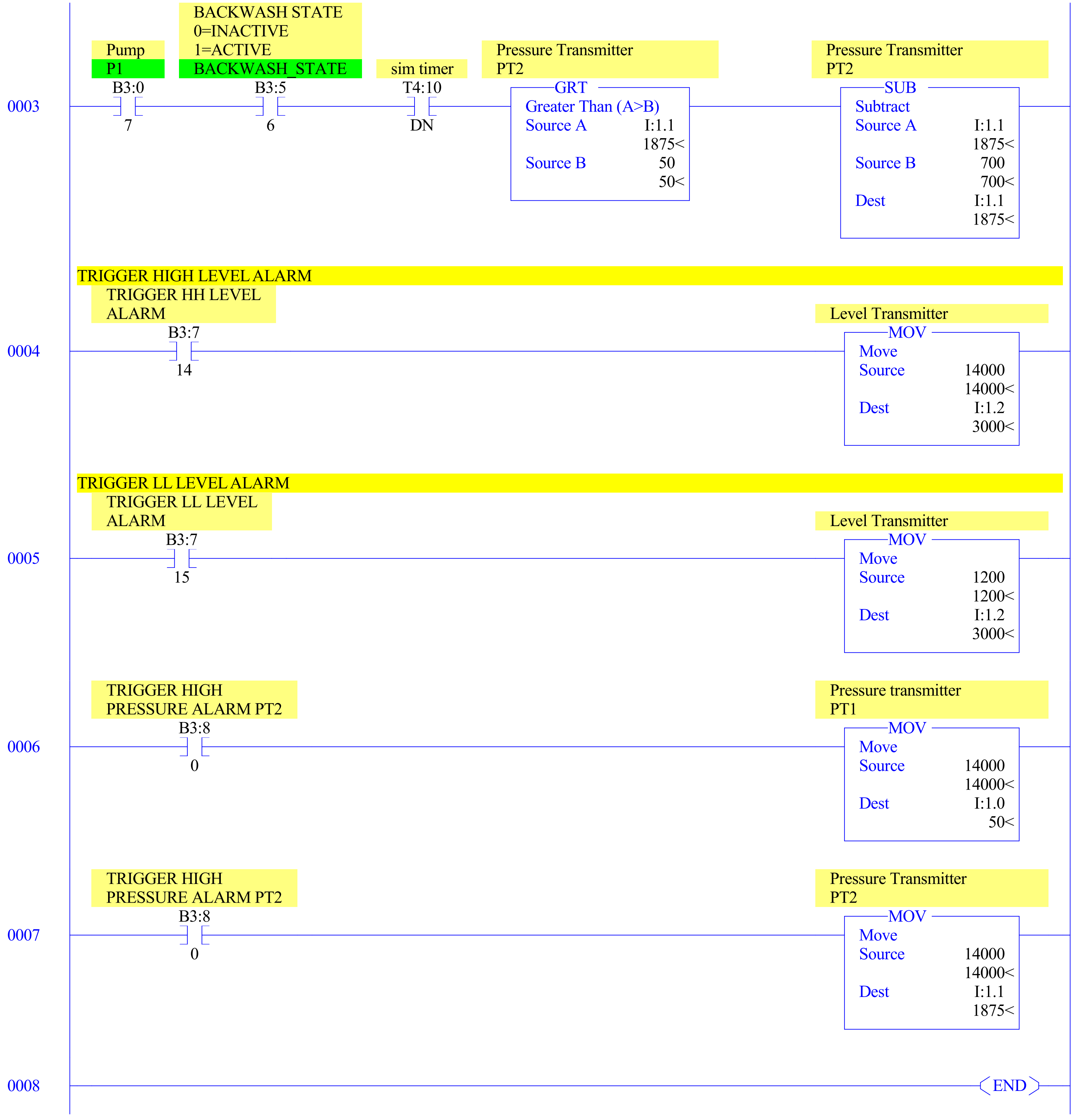

0013
BACKWASH LOCKOUT STATE
0=NOT LOCKED OUT
1=LOCKED OUT
BW_LO_STATE
B3:7
2
Enable backwash
BW_ENABLE_PB
B3:7
3
one
B3:7
ONS
4
Backwash Enable Trigger
B3:7
5
0014
Backwash lockout trigger
B3:7
1
BACKWASH LOCKOUT STATE
0=NOT LOCKED OUT
1=LOCKED OUT
BW_LO_STATE
B3:7
2
Backwash Enable Trigger
B3:7
5
BACKWASH LOCKOUT STATE
0=NOT LOCKED OUT
1=LOCKED OUT
BW_LO_STATE
B3:7
2
0015
END


LOW FLOW ALARM AND NOTIFICATION
0000
Flow switch FS-1
FS1
B3:0
0
Pump
P1
B3:0
7
ALARM RESET
ALARM_RESET
B3:5
15
Flow alarm activation timer
TON
Timer On Delay
Timer T4:6
Time Base 1.0
Preset 5<
Accum 0<
EN
DN
0001
Flow alarm activation timer
T4:6
DN
one
B3:5
ONS
12
Flow alarm trigger
B3:5
13
0002
Flow alarm trigger
B3:5
13
Flow alarm
B3:5
14
Flow alarm
B3:5
14
ALARM RESET
ALARM_RESET
B3:5
15
0003
Flow alarm trigger
B3:5
13
Flow alarm notfication
FLOW_ALM_NOTIFY
B3:6
1
Flow alarm notfication
FLOW_ALM_NOTIFY
B3:6
1
ALARM RESET
ALARM_RESET
B3:5
15
ALARM SILENCE
ALARM_SILENCE
B3:6
0
HH LEVEL TANK ALARM AND NOTIFICATION
0004
Tank Level Scaled
LT1
GRT
Greater Than (A>B)
Source A F8:3
18.31166<
Source B F8:7
90.0<
ALARM RESET
ALARM_RESET
B3:5
15
HH tank level alarm activation timer
TON
Timer On Delay
Timer T4:7
Time Base 1.0
Preset 10<
Accum 0<
EN
DN
0005
HH tank level alarm activation timer
T4:7
DN
one
B3:6
ONS
2
HH level alarm trigger
B3:6
3

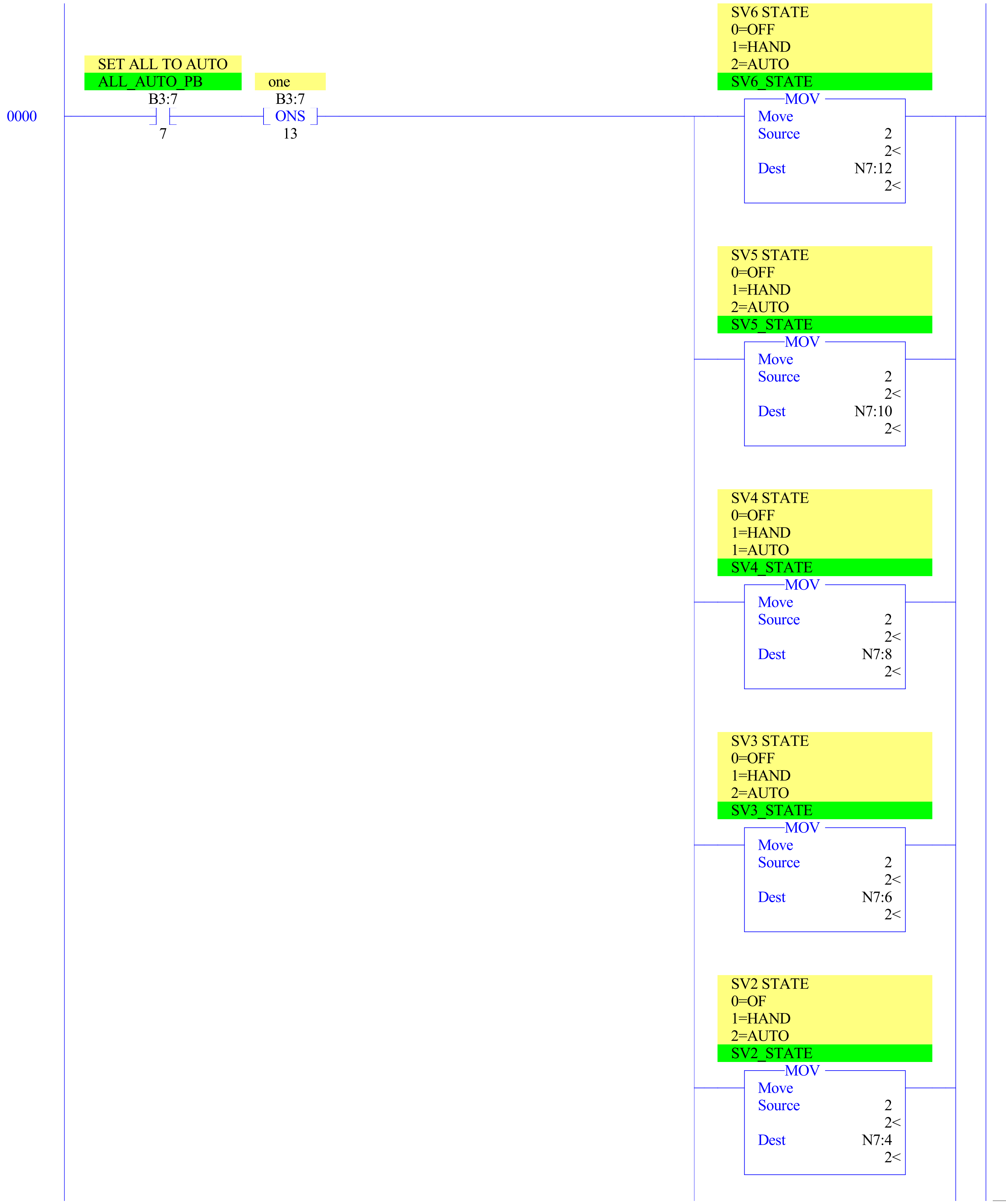

0006
HH level alarm trigger
B3:6
3
HH LEVEL ALARM
B3:6
4
HH LEVEL ALARM
B3:6
4
ALARM RESET
ALARM_RESET
B3:5
15
0007
HH level alarm trigger
B3:6
3
HH level alarm notify
HH_LEVEL_NOTIFY
B3:6
5
HH level alarm notify
HH_LEVEL_NOTIFY
B3:6
5
ALARM RESET
ALARM_RESET
B3:5
15
ALARM SILENCE
ALARM_SILENCE
B3:6
0
LL LEVEL TANK ALARM AND NOTIFICATION
0008
Tank Level Scaled
LT1
LES
Less Than (A<B)
Source A F8:3
18.31166<
Source B F8:8
15.0<
ALARM RESET
ALARM_RESET
B3:5
15
LL Tank level alarm trigger
TON
Timer On Delay
Timer T4:8
Time Base 1.0
Preset 10<
Accum 0<
EN
DN
0009
LL Tank level alarm trigger
T4:8
DN
one
B3:6
ONS
6
LL tank alarm trigger
B3:6
7
0010
LL tank alarm trigger
B3:6
7
LL Level alarm
B3:6
8
LL Level alarm
B3:6
8
ALARM RESET
ALARM_RESET
B3:5
15
0011
LL tank alarm trigger
B3:6
7
LL alarm notify
LL_LEVEL_NOTIFY
B3:6
9
LL alarm notify
LL_LEVEL_NOTIFY
B3:6
9
ALARM RESET
ALARM_RESET
B3:5
15
ALARM SILENCE
ALARM_SILENCE
B3:6
0

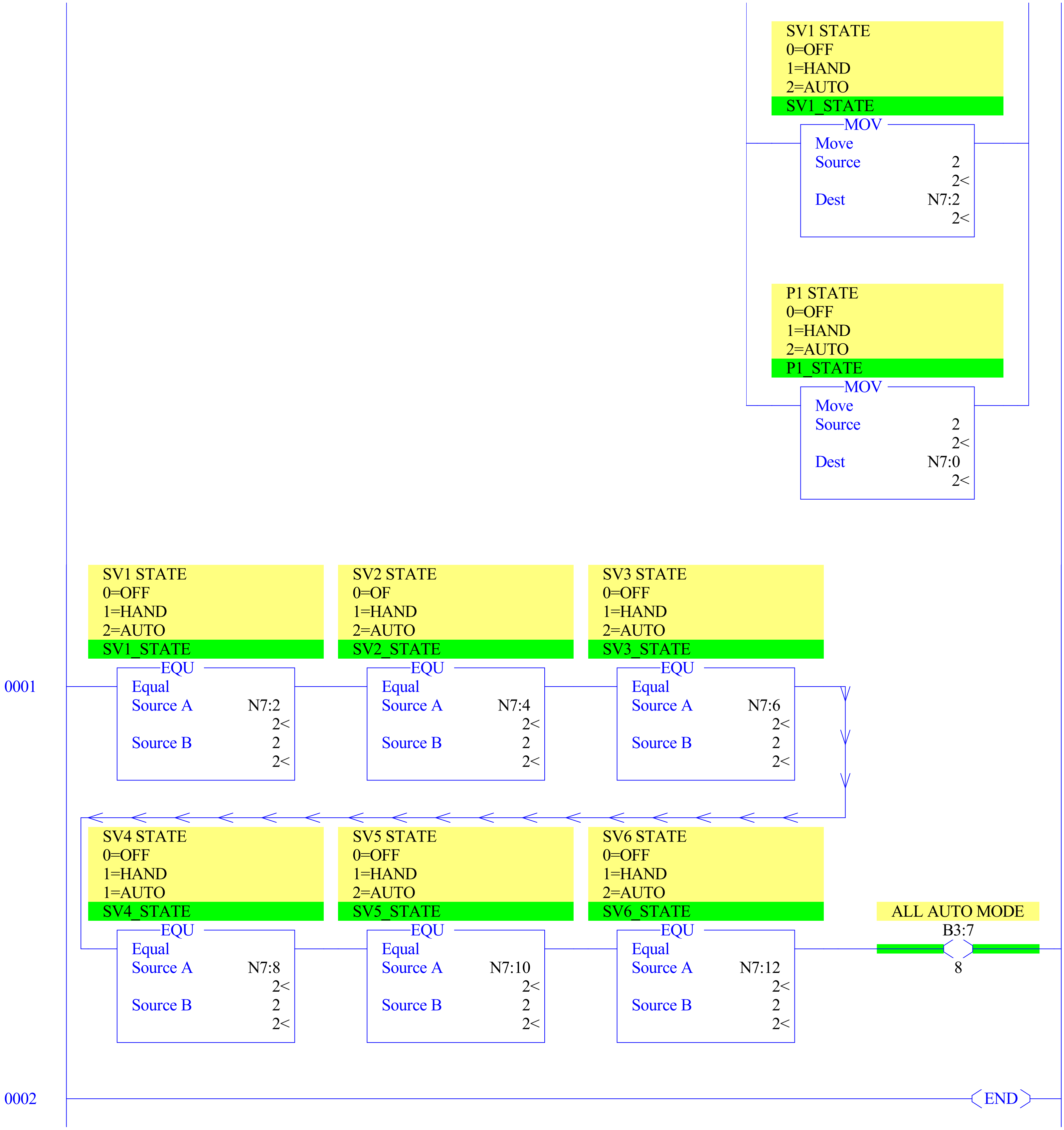

HIGH FILTER PRESSURE ALARM AND NOTIFICATION
0012
Pressure PT1
PT1
GRT
Greater Than (A>B)
Source A F8:0
0.06103888<
Source B F8:9
15.0<
ALARM RESET
ALARM_RESET
B3:5
15
High filter pressure
alarm
trigger timer
TON
Timer On Delay
Timer T4:9
Time Base 1.0
Preset 10<
Accum 0<
EN
DN
Pressure PT2
PT2
GRT
Greater Than (A>B)
Source A F8:1
2.288958<
Source B F8:9
15.0<
0013
High filter pressure
alarm
trigger timer
T4:9
DN
one
B3:6
ONS
10
High Filter Pressure
alarm trigger
B3:6
11
0014
High Filter Pressure
alarm trigger
B3:6
11
High Filter Pressure
Alarm
B3:6
12
High Filter Pressure
Alarm
B3:6
12
ALARM RESET
ALARM_RESET
B3:5
15
0015
High Filter Pressure
alarm trigger
B3:6
11
High pressure alarm
notify
H_FILTER_PRES_NOTIFY
B3:6
13
High pressure alarm
notify
H_FILTER_PRES_NOTIFY
B3:6
13
ALARM RESET
ALARM_RESET
B3:5
15
ALARM SILENCE
ALARM_SILENCE
B3:6
0
0016
END


0000
sim timer
T4:10
DN
sim timer
TON
Timer On Delay
Timer T4:10
Time Base 1.0
Preset 3<
Accum 0<
EN
DN
SIMULATED TANK LEVEL AND FILTER PRESSURES RISING AND FALLING FOR VISULATISATION AND TESTING PURPOSES
0001
Pump
P1
B3:0
7
sim timer
T4:10
DN
BACKWASH STATE
0=INACTIVE
1=ACTIVE
BACKWASH_STATE
B3:5
6
Level Transmitter
ADD
Add
Source A 250
250<
Source B I:1.2
3000<
Dest I:1.2
3000<
Pressure Transmitter
PT2
ADD
Add
Source A I:1.1
1875<
Source B 25
25<
Dest I:1.1
1875<
0002
SYSTEM STATE
0=OFF
1=ON
SYSTEM_STATE
B3:4
7
DISCHARGE VALVE
DISCHARGE_VALVE
B3:7
12
Pump
P1
B3:0
7
BACKWASH STATE
0=INACTIVE
1=ACTIVE
BACKWASH_STATE
B3:5
6
Tank Level Scaled
LT1
GRT
Greater Than (A>B)
Source A F8:3
18.31166<
Source B 20.0
20.0<
sim timer
T4:10
DN
Level Transmitter
SUB
Subtract
Source A I:1.2
3000<
Source B 500
500<
Dest I:1.2
3000<


0003
Pump
P1
B3:0
7
BACKWASH STATE
0=INACTIVE
1=ACTIVE
BACKWASH_STATE
B3:5
6
sim timer
T4:10
DN
Pressure Transmitter
PT2
GRT
Greater Than (A>B)
Source A I:1.1
1875<
Source B 50
50<
Pressure Transmitter
PT2
SUB
Subtract
Source A I:1.1
1875<
Source B 700
700<
Dest I:1.1
1875<
TRIGGER HIGH LEVEL ALARM
0004
TRIGGER HH LEVEL
ALARM
B3:7
14
Level Transmitter
MOV
Move
Source 14000
14000<
Dest I:1.2
3000<
TRIGGER LL LEVEL ALARM
0005
TRIGGER LL LEVEL
ALARM
B3:7
15
Level Transmitter
MOV
Move
Source 1200
1200<
Dest I:1.2
3000<
0006
TRIGGER HIGH
PRESSURE ALARM PT2
B3:8
0
Pressure transmitter
PT1
MOV
Move
Source 14000
14000<
Dest I:1.0
50<
0007
TRIGGER HIGH
PRESSURE ALARM PT2
B3:8
0
Pressure Transmitter
PT2
MOV
Move
Source 14000
14000<
Dest I:1.1
1875<
0008
END

LAD 10 - SETTOAUTO --- Total Rungs in File = 3




SV1 STATE
0=OFF
1=HAND
2=AUTO
SV1_STATE
MOV
Move
Source 2
2<
Dest N7:2
2<
P1 STATE
0=OFF
1=HAND
2=AUTO
P1_STATE
MOV
Move
Source 2
2<
Dest N7:0
2<
0001
SV1 STATE
0=OFF
1=HAND
2=AUTO
SV1_STATE
EQU
Equal
Source A N7:2
2<
Source B 2
2<
SV2 STATE
0=OF
1=HAND
2=AUTO
SV2_STATE
EQU
Equal
Source A N7:4
2<
Source B 2
2<
SV3 STATE
0=OFF
1=HAND
2=AUTO
SV3_STATE
EQU
Equal
Source A N7:6
2<
Source B 2
2<
SV4 STATE
0=OFF
1=HAND
1=AUTO
SV4_STATE
EQU
Equal
Source A N7:8
2<
Source B 2
2<
SV5 STATE
0=OFF
1=HAND
2=AUTO
SV5_STATE
EQU
Equal
Source A N7:10
2<
Source B 2
2<
SV6 STATE
0=OFF
1=HAND
2=AUTO
SV6_STATE
EQU
Equal
Source A N7:12
2<
Source B 2
2<
ALL AUTO MODE
B3:7
8
0002
END

Data File O0 (bin) -- OUTPUT

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
O:0.0	0	0	0	0	0	0	0	1	0	1	0	1	0	1	0	0	Bul.1763	MicroLogix 1100 Series B
O:0.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763	MicroLogix 1100 Series B
O:0.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763	MicroLogix 1100 Series B
O:0.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763	MicroLogix 1100 Series B

Data File I1 (bin) -- INPUT

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
I:0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763 MicroLogix 1100 Series B
I:0.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763 MicroLogix 1100 Series B
I:0.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763 MicroLogix 1100 Series B
I:0.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763 MicroLogix 1100 Series B
I:0.4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763 MicroLogix 1100 Series B-Analog
I:0.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763 MicroLogix 1100 Series B-Analog
I:1.0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	1762-IF4 - Analog 4 Chan. Input
I:1.1	0	0	0	0	0	1	1	1	0	1	0	1	0	0	1	1	1762-IF4 - Analog 4 Chan. Input
I:1.2	0	0	0	0	1	0	1	1	1	0	1	1	1	0	0	0	1762-IF4 - Analog 4 Chan. Input
I:1.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1762-IF4 - Analog 4 Chan. Input
I:1.4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1762-IF4 - Analog 4 Chan. Input
I:1.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1762-IF4 - Analog 4 Chan. Input
I:1.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1762-IF4 - Analog 4 Chan. Input

Main

Processor Mode S:1/0 - S:1/4 = Remote Program Mode
On Power up Go To Run (Mode Behavior) S:1/12 = 0
First Pass S:1/15 = No
Free Running Clock S:4 = 0010-0010-0010-1101

Proc

OS Catalog Number S:57 = 1100
OS Series S:58 = B
OS FRS S:59 =
Processor Catalog Number S:60 =
Processor Series S:61 = A
Processor FRN S:62 =

User Program Type S:63 = 8001h
Compiler Revision Number S:64 =

Scan Times

Maximum (x10 ms) S:22 = 20
Watchdog (x10 ms) S:3 (high byte) = 10
Last 100 uSec Scan Time S:35 = 0
Scan Toggle Bit S:33/9 = 0

Math

Math Overflow Selected S:2/14 = 0
Overflow Trap S:5/0 = 0
Carry S:0/0 = 0
Overflow S:0/1 = 0
Zero Bit S:0/2 = 0
Sign Bit S:0/3 = 0

Math Register (lo word) S:13 = 0
Math Register (high word) S:14-S:13 = 0
Math Register (32 Bit) S:14-S:13 = 0

Chan 0

Processor Mode S:1/0- S:1/4 = Remote Program Mode
Node Address S:15 (low byte) = 0
Baud Rate S:15 (high byte) = ?
Channel Mode S:33/3 = 0
Comms Active S:33/4 = 0
Incoming Cmd Pending S:33/0 = 0
Msg Reply Pending S:33/1 = 0

Outgoing Msg Cmd Pending S:33/2 = 0

Debug

Suspend Code S:7 = 0
Suspend File S:8 = 0

Errors

Fault Override At Power Up S:1/8 = 0
Startup Protection Fault S:1/9 = 0
Major Error Halt S:1/13 = 0
Overflow Trap S:5/0 = 0
Control Register Error S:5/2 = 0
Major Error Executing User Fault Rtn. S:5/3 = 0
Battery Low S:5/11 = 0
Input Filter Selection Modified S:5/13 = 0
ASCII String Manipulation error S:5/15 = 0

Fault Routine S:29 = 0
Major Error S:6 = 0h
Error Description:

Protection

Deny Future Access S:1/14 = No
Data File Overwrite Protection Lost S:36/10 = False

Mem Module

Memory Module Loaded On Boot S:5/8 = 0
Password Mismatch S:5/9 = 0
Load Memory Module On Memory Error S:1/10 = 0
Load Memory Module Always S:1/11 = 0
On Power up Go To Run (Mode Behavior) S:1/12 = 0
Program Compare S:2/9 = 0
Data File Overwrite Protection Lost S:36/10 = 0

Forces

Forces Enabled S:1/5 = Yes
Forces Installed S:1/6 = No

Data File B3 (bin) -- BINARY

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	(Symbol) Description
B3:0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	0	
B3:1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B3:2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B3:3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B3:4	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	
B3:5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B3:6	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	
B3:7	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	
B3:8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Data File T4 -- TIMER

Offset	EN	TT	DN	BASE	PRE	ACC	(Symbol) Description
T4:0	0	0	0	1.0 sec	10	0	Tank high level timer
T4:1	1	0	1	1.0 sec	10	10	Tank low level timer
T4:2	1	1	0	1.0 sec	60	22	Runtime-seconds
T4:3	0	0	0	1.0 sec	10	0	Pending backwash start timer
T4:4	0	0	0	1.0 sec	60	0	Backwash timer
T4:5	0	0	0	1.0 sec	5	0	BW COUNT RESET PB TIMER
T4:6	0	0	0	1.0 sec	5	0	Flow alarm activation timer
T4:7	0	0	0	1.0 sec	10	0	HH tank level alarm activation timer
T4:8	0	0	0	1.0 sec	10	0	LL Tank level alarm trigger
T4:9	0	0	0	1.0 sec	10	0	High filter pressure alarm trigger timer
T4:10	1	1	0	1.0 sec	3	0	sim timer

Data File C5 -- COUNTER

Offset	CU	CD	DN	OV	UN	UA	PRE	ACC	(Symbol) Description
C5:0	0	0	0	0	0	0	0	0	

Data File R6 -- CONTROL

Offset	EN	EU	DN	EM	ER	UL	IN	FD	LEN	POS	(Symbol) Description
R6:0	0	0	0	0	0	0	0	0	0	0	

Data File N7 (dec) -- INTEGER

Offset	0	1	2	3	4	5	6	7	8	9
N7:0	2	2	2	2	2	2	2	2	2	0
N7:10	2	2	2	0	41	0	0	0	60	

Data File F8 -- FLOAT

Offset	0	1	2	3	4
F8:0	0.06103888	2.288958	0	18.31166	-2.227919
F8:5	2.227919	12	90	15	15
F8:10	80	20	0	0	0

AUTOMATED WATER STORAGE & FILTRATION CONTROL SYSTEM

Address/Symbol Database

Address	Symbol	Scope	Description	Sym Group	Dev. Code
B3:0/0	FS1	Global	Flow switch FS-1		
B3:0/1	SV1	Global	SV1		
B3:0/2	SV2	Global	SV2		
B3:0/3	SV3	Global	SV3		
B3:0/4	SV4	Global	SV4 ip		
B3:0/5	SV5	Global	SV5		
B3:0/6	SV6	Global	SV6		
B3:0/7	P1	Global	Pump		
B3:0/8	P1_HAND_PB	Global	P1 HAND PB		
B3:0/9			one		
B3:0/10			one		
B3:0/11			one		
B3:0/12			one		
B3:0/13	P1_AUTO_PB	Global	P1 AUTO PB		
B3:0/14			one		
B3:0/15	P1_OFF_PB	Global	P1 OFF PB		
B3:1/0			one		
B3:1/1	SV1_HAND_PB	Global	SV1 HAND PB		
B3:1/2			one		
B3:1/3			one		
B3:1/4			one		
B3:1/5			one		
B3:1/6	SV1_AUTO_PB	Global	SV1 AUTO PB		
B3:1/7			one		
B3:1/8	SV1_OFF_PB	Global	SV1 OFF PB		
B3:1/9	SV2_HAND_PB	Global	SV2 HAND PB		
B3:1/10			one		
B3:1/11			one		
B3:1/12			one		
B3:1/13			one		
B3:1/14	SV2_AUTO_PB	Global	SV2 AUTO PB		
B3:1/15			one		
B3:2/0	SV2_OFF_PB	Global	SV2 OFF PB		
B3:2/1			one		
B3:2/2	SV3_HAND_PB	Global	SV3 HAND PB		
B3:2/3			one		
B3:2/4			one		
B3:2/5			one		
B3:2/6			one		
B3:2/7	SV3_AUTO_PB	Global	SV3 AUTO PB		
B3:2/8			one		
B3:2/9	SV3_OFF_PB	Global	SV3 OFF PB		
B3:2/10			one		
B3:2/11	SV4_HAND_PB	Global	SV4 HAND PB		
B3:2/12			one		
B3:2/13			one		
B3:2/14			one		
B3:2/15			one		
B3:3/0	SV4_AUTO_PB	Global	SV4 AUTO PB		
B3:3/1			one		
B3:3/2	SV4_OFF_PB	Global	SV4 OFF PB		
B3:3/3			one		
B3:3/4	SV5_HAND_PB	Global	SV5 HAND PB		
B3:3/5			one		
B3:3/6			one		
B3:3/7			one		
B3:3/8			one		
B3:3/9	SV5_AUTO_PB	Global	SV5 AUTO PB		
B3:3/10			one		
B3:3/11	SV5_OFF_PB	Global	SV5 OFF PB		
B3:3/12			one		
B3:3/13	SV6_HAND_PB	Global	SV6 HAND PB		
B3:3/14			one		
B3:3/15			one		
B3:4/0			one		
B3:4/1			one		
B3:4/2	SV6_AUTO_PB	Global	SV6 AUTO PB		
B3:4/3			one		
B3:4/4	SV6_OFF_PB	Global	SV6 OFF PB		
B3:4/5			one		
B3:4/6			one		
B3:4/7	SYSTEM_STATE	Global	SYSTEM STATE 0=OFF 1=ON		
B3:4/8			PUMP START TRIGGER		
B3:4/9			one		
B3:4/10			PUMP OFF TRIGGER		
B3:4/11			PUMP ON		
B3:4/12	SYSTEM_START_PB	Global	SYSTEM START PB		
B3:4/13			one		
B3:4/15			SYSTEM STARTER		
B3:5/0	SYSTEM_STOP_PB	Global	SYSTEM STOP PB		
B3:5/1			one		
B3:5/2			SYSTEM STOPPER		
B3:5/3			Pending backwash trigger		
B3:5/4			one		

Address/Symbol Database

Address	Symbol	Scope	Description	Sym Group	Dev. Code
B3:5/5	BACKWASH_PENDING	Global	Pending backwash		
B3:5/6	BACKWASH_STATE	Global	BACKWASH STATE 0=INACTIVE 1=ACTIVE		
B3:5/7			one		
B3:5/8			one		
B3:5/9	BW_COUNT_RESET_PB	Global	Backwash Counht rest PB		
B3:5/10			one		
B3:5/11	BW_LOG_TRIGGER	Global	Backwash End Data Log Trigger		
B3:5/12			one		
B3:5/13			Flow alarm trigger		
B3:5/14			Flow alarm		
B3:5/15	ALARM_RESET	Global	ALARM RESET		
B3:6/0	ALARM_SILENCE	Global	ALARM SILENCE		
B3:6/1	FLOW_ALM_NOTIFY	Global	Flow alarm notficication		
B3:6/2			one		
B3:6/3			HH level alarm trigger		
B3:6/4			HH LEVEL ALARM		
B3:6/5	HH_LEVEL_NOTIFY	Global	HH level alarm notify		
B3:6/6			one		
B3:6/7			LL tank alarm trigger		
B3:6/8			LL Level alarm		
B3:6/9	LL_LEVEL_NOTIFY	Global	LL alarm notify		
B3:6/10			one		
B3:6/11			High Filter Pressure alarm trigger		
B3:6/12			High Filter Pressure Alarm		
B3:6/13	H_FILTER_PRES_NOTIFY	Global	High pressure alarm notify		
B3:6/14	BW_LOCKOUT_PB	Global	Lockout Backwash PB		
B3:7/0			one		
B3:7/1			Backwash lockout trigger		
B3:7/2	BW_LO_STATE	Global	BACKWASH LOCKOUT STATE 0=NOT LOCKED OUT 1=LOCKED OUT		
B3:7/3	BW_ENABLE_PB	Global	Enable backwash		
B3:7/4			one		
B3:7/5			Backwash Enable Trigger		
B3:7/6	MANUAL_BW_PB	Global	MANUAL BACKWASH PUSHBUTTON		
B3:7/7	ALL_AUTO_PB	Global	SET ALL TO AUTO		
B3:7/8			ALL AUTO MODE		
B3:7/9	RUNTIME_RESET_PB	Global	Runtime Reset PB		
B3:7/10			one		
B3:7/11			Runtime reset trigger		
B3:7/12	DISCHARGE_VALVE	Global	DISCHARGE VALVE		
B3:7/13			one		
B3:7/14			TRIGGER HH LEVEL ALARM		
B3:7/15			TRIGGER LL LEVEL ALARM		
B3:8/0			TRIGGER HIGH PRESSURE ALARM PT2		
F8:0	PT1	Global	Pressure PT1		
F8:1	PT2	Global	Pressure PT2		
F8:2					
F8:3	LT1	Global	Tank Level Scaled		
F8:4			Diff. pressure		
F8:5	DIFF_PRESS_ABS	Global	Diff Pressure- absolute		
F8:6	DIFF_PRES_SP	Global	Max. Diff Pressue (set from HMI)		
F8:7	HH_LEVEL_SP	Global	HH Tank level (set on the HMI)		
F8:8	LL_LEVEL_SP	Global	LL level (set from HMI)		
F8:9	H_PRES_SP	Global	High pressure (set from HMI)		
F8:10	H_LEVEL	Global	H_TANK_LEVEL		
F8:11	L_LEVEL	Global	L_TANK_LEVEL		
F8:30					
I:0/0			SV1 ip		
I:0/1			SV1 ip		
I:0/2			SV2 ip		
I:0/3			SV3 ip		
I:0/4			SV4 ip		
I:0/5			SV5 ip		
I:0/6			SV6 ip		
I:1.0			Pressure transmitter PT1		
I:1/0			Pressure transmitter PT1 (psi)		
I:1.1			Pressure Transmitter PT2		
I:1.2			Level Transmitter		
N7:0	P1_STATE	Global	P1 STATE 0=OFF 1=HAND 2=AUTO		
N7:1			P1 state temporary storage		
N7:2	SV1_STATE	Global	SV1 STATE 0=OFF 1=HAND 2=AUTO		
N7:3			SV1 temporary storage bit		
N7:4	SV2_STATE	Global	SV2 STATE 0=OF 1=HAND 2=AUTO		
N7:5			SV2 temporary storage bit		
N7:6	SV3_STATE	Global	SV3 STATE 0=OFF 1=HAND 2=AUTO		
N7:7			SV3 state temporary storage bit		
N7:8	SV4_STATE	Global	SV4 STATE 0=OFF 1=HAND 1=AUTO		
N7:9			SV4 State temporary storage bit		
N7:10	SV5_STATE	Global	SV5 STATE 0=OFF 1=HAND 2=AUTO		
N7:11			SV5 state temporary storage bit		
N7:12	SV6_STATE	Global	SV6 STATE 0=OFF 1=HAND 2=AUTO		
N7:13			SV6 state temporary storage bit		
N7:14	MINUTES	Global	Runtime-minutes		
N7:15	HOURS	Global	Runtime-hours		
N7:16	THOUSAND_HOURS	Global	Runtime-thousands of hours		

Address	Symbol	Scope	Description	Sym Group	Dev. Code
N7:17	BACKWASH_COUNT	Global	Backwash Cycle Counter		
N7:18	BW_TIME_LEFT	Global	Backwash time left		
O:0/0			Pump OP		
O:0/1			FS-1		
O:0/2			SV1		
O:0/3			SV2		
O:0/4			SV3		
O:0/5			SV5		
O:0/6			SV5		
O:0/7			SV6		
O:0/8			DISCHARGE VALVE		
S:0			Arithmetic Flags		
S:0/0			Processor Arithmetic Carry Flag		
S:0/1			Processor Arithmetic Underflow/ Overflow Flag		
S:0/2			Processor Arithmetic Zero Flag		
S:0/3			Processor Arithmetic Sign Flag		
S:1			Processor Mode Status/ Control		
S:1/0			Processor Mode Bit 0		
S:1/1			Processor Mode Bit 1		
S:1/2			Processor Mode Bit 2		
S:1/3			Processor Mode Bit 3		
S:1/4			Processor Mode Bit 4		
S:1/5			Forces Enabled		
S:1/6			Forces Present		
S:1/7			Comms Active		
S:1/8			Fault Override at Powerup		
S:1/9			Startup Protection Fault		
S:1/10			Load Memory Module on Memory Error		
S:1/11			Load Memory Module Always		
S:1/12			Load Memory Module and RUN		
S:1/13			Major Error Halted		
S:1/14			Access Denied		
S:1/15			First Pass		
S:2/0			STI Pending		
S:2/1			STI Enabled		
S:2/2			STI Executing		
S:2/3			Index Addressing File Range		
S:2/4			Saved with Debug Single Step		
S:2/5			DH-485 Incoming Command Pending		
S:2/6			DH-485 Message Reply Pending		
S:2/7			DH-485 Outgoing Message Command Pending		
S:2/15			Comms Servicing Selection		
S:3			Current Scan Time/ Watchdog Scan Time		
S:4			Time Base		
S:5/0			Overflow Trap		
S:5/2			Control Register Error		
S:5/3			Major Err Detected Executing UserFault Routine		
S:5/4			M0-M1 Referenced on Disabled Slot		
S:5/8			Memory Module Boot		
S:5/9			Memory Module Password Mismatch		
S:5/10			STI Overflow		
S:5/11			Battery Low		
S:6			Major Error Fault Code		
S:7			Suspend Code		
S:8			Suspend File		
S:9			Active Nodes		
S:10			Active Nodes		
S:11			I/O Slot Enables		
S:12			I/O Slot Enables		
S:13			Math Register		
S:14			Math Register		
S:15			Node Address/ Baud Rate		
S:16			Debug Single Step Rung		
S:17			Debug Single Step File		
S:18			Debug Single Step Breakpoint Rung		
S:19			Debug Single Step Breakpoint File		
S:20			Debug Fault/ Powerdown Rung		
S:21			Debug Fault/ Powerdown File		
S:22			Maximum Observed Scan Time		
S:23			Average Scan Time		
S:24			Index Register		
S:25			I/O Interrupt Pending		
S:26			I/O Interrupt Pending		
S:27			I/O Interrupt Enabled		
S:28			I/O Interrupt Enabled		
S:29			User Fault Routine File Number		
S:30			STI Setpoint		
S:31			STI File Number		
S:32			I/O Interrupt Executing		
S:33			Extended Proc Status Control Word		
S:33/0			Incoming Command Pending		
S:33/1			Message Reply Pending		
S:33/2			Outgoing Message Command Pending		
S:33/3			Selection Status User/DF1		

Address	Symbol	Scope	Description	Sym Group	Dev. Code
S:33/4			Communicat Active		
S:33/5			Communicat Servicing Selection		
S:33/6			Message Servicing Selection Channel 0		
S:33/7			Message Servicing Selection Channel 1		
S:33/8			Interrupt Latency Control Flag		
S:33/9			Scan Toggle Flag		
S:33/10			Discrete Input Interrupt Reconfigur Flag		
S:33/11			Online Edit Status		
S:33/12			Online Edit Status		
S:33/13			Scan Time Timebase Selection		
S:33/14			DTR Control Bit		
S:33/15			DTR Force Bit		
S:34			Pass-thru Disabled		
S:34/0			Pass-Thru Disabled Flag		
S:34/1			DH+ Active Node Table Enable Flag		
S:34/2			Floating Point Math Flag Disable,Fl		
S:35			Last 1 ms Scan Time		
S:36			Extended Minor Error Bits		
S:36/8			DII Lost		
S:36/9			STI Lost		
S:36/10			Memory Module Data File Overwrite Protection		
S:37			Clock Calendar Year		
S:38			Clock Calendar Month		
S:39			Clock Calendar Day		
S:40			Clock Calendar Hours		
S:41			Clock Calendar Minutes		
S:42			Clock Calendar Seconds		
S:43			STI Interrupt Time		
S:44			I/O Event Interrupt Time		
S:45			DII Interrupt Time		
S:46			Discrete Input Interrupt- File Number		
S:47			Discrete Input Interrupt- Slot Number		
S:48			Discrete Input Interrupt- Bit Mask		
S:49			Discrete Input Interrupt- Compare Value		
S:50			Processor Catalog Number		
S:51			Discrete Input Interrupt- Return Number		
S:52			Discrete Input Interrupt- Accumulat		
S:53			Reserved/ Clock Calendar Day of the Week		
S:55			Last DII Scan Time		
S:56			Maximum Observed DII Scan Time		
S:57			Operating System Catalog Number		
S:58			Operating System Series		
S:59			Operating System FRN		
S:61			Processor Series		
S:62			Processor Revision		
S:63			User Program Type		
S:64			User Program Functional Index		
S:65			User RAM Size		
S:66			Flash EEPROM Size		
S:67			Channel 0 Active Nodes		
S:68			Channel 0 Active Nodes		
S:69			Channel 0 Active Nodes		
S:70			Channel 0 Active Nodes		
S:71			Channel 0 Active Nodes		
S:72			Channel 0 Active Nodes		
S:73			Channel 0 Active Nodes		
S:74			Channel 0 Active Nodes		
S:75			Channel 0 Active Nodes		
S:76			Channel 0 Active Nodes		
S:77			Channel 0 Active Nodes		
S:78			Channel 0 Active Nodes		
S:79			Channel 0 Active Nodes		
S:80			Channel 0 Active Nodes		
S:81			Channel 0 Active Nodes		
S:82			Channel 0 Active Nodes		
S:83			DH+ Active Nodes		
S:84			DH+ Active Nodes		
S:85			DH+ Active Nodes		
S:86			DH+ Active Nodes		
T4:0			Tank high level timer		
T4:0/DN					
T4:1			Tank low level timer		
T4:1/DN					
T4:2			Runtime-seconds		
T4:2.ACC	SECONDS	Global			
T4:2/DN					
T4:3			Pending backwash start timer		
T4:3/DN					
T4:4			Backwash timer		
T4:4.ACC					
T4:4/DN					
T4:4/TT					
T4:5			BW COUNT RESET PB TIMER		
T4:5/DN					

Address/Symbol Database

Address	Symbol	Scope	Description	Sym Group	Dev. Code
T4:6			Flow alarm activation timer		
T4:6/DN					
T4:7			HH tank level alarm activation timer		
T4:7/DN					
T4:8			LL Tank level alarm trigger		
T4:8/DN					
T4:9			High filter pressure alarm trigger timer		
T4:9/DN					
T4:10			sim timer		
T4:10/DN					
U:3			I/O		
U:4			HOA		
U:5			CONTROLS		
U:6			RUNTIME		
U:7			BACKWASH		
U:8			ALARM		
U:9			SIMULATION		
U:10			SETTOAUTO		

Address	Instruction	Description

Symbol Group Database

Group_Name	Description